Dispersion modelling of gaseous air pollutants from the Digboi oil refinery

M.Tech Thesis

Submitted in partial fulfillment of the requirements
for the degree of

Master of Technology
In
Environmental Engineering
by

Sanjay Adhikary
Roll no : 204104115

Supervisor: **Prof. Sharad Gokhale**



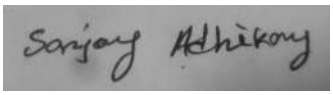
Department of Civil Engineering
Indian Institute of Technology Guwahati
Guwahati-781039

July, 2022

Indian Institute of Technology Guwahati
Department of Civil Engineeing

DECLARATION

I, hereby declare that the project titled **"Dispersion modelling of gaseous air pollutants from the Digboi oil refinery area"** submitted herein for the requirements of the degree of Master of Technology in Environmental Engineering has been carried out by me in the Department of Civil Engineering of Indian Institute of Technology Guwahati during a period from July 2021 to July 2022. The work is original and has not been submitted as a whole or part thereof for the award of degree at this or any other institution.



July, 2022
Sanjay Adhikary
Roll no:204104115
Department of Civil Engineering,
Indian Institute of Technology
Guwahati-781039, India.

Indian Institute of Technology Guwahati
Department of Civil Engineeing

CERTIFICATE

This is to certify that the research work presented in this thesis entitled **"Dispersion modellig of gaseous air pollutants from the Digboi oil refinery area"** is a bone fide work carried out by **SANJAY ADHIKARY**(Roll no:204104115) in partial fulfillment for the award of **Master of Technology in Environmental Engineering** of the Indian Institute of Technology, Guwahati during the year 2021-2022. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the thesis deposited in the department library.The thesis report has been approved as it is satisfied the academic requirements in respect of thesis work prescribed for the said Degree.



July, 2022

Supervisor: **Prof.Sharad Gokhale**
Department of Civil Engineering,
Indian Institute of Technology
Guwahati-781039, India.

ACKNOWLEDGEMENT

It is quite difficult to express my gratitude in a few lines to all those who have assisted me in completing my thesis. Many people assisted me in many ways throughout this learning journey. I hope those I've listed understand that my gratitude stretches far beyond the next paragraphs.

To begin, I'd like to offer my sincere and heartfelt gratitude to my supervisor, Professor Sharad B. Gokhale, Department of Civil Engineering, IIT Guwahati, for his advice, support, and consistent inspiration during my project work, without which this project would not have been possible. My supervisor, who has kept faith in me throughout my time at IIT Guwahati, is responsible for all of my achievements. I am grateful for the time he has given me for meetings about thoughtful ideas and discussions about my research work, which has helped me strengthen my subject matter expertise and forced me to think outside the box.

I would also like to acknowledge my sincere gratitude to other faculty members of Department of Civil Engineering, Professor Saswati Chakraborty, Proffesor M. Jawed, Professor Pranab Kumar Ghosh and Professor Ajay Kalamdhad for their helpful suggestions during my entire course work. Again, I thank my guide as well as the Head of the Department of Civil Engineering, Professor Sharad Gokhale for providing me all the facilities necessary for carrying out my research work.

I am thankful to my seniors Mr. Argha K Guha, Mr. Neeldip Barman, Mr. Sameer Singh, Mr. Rajarshi Sharma, Mr. Mukesh Tiwari, Mr. Himanshu Gupta and Ms. Rashmi Devi for their valuable suggestions and guidance which helped me in completing my project work. A special thanks to Mr. Lavanya, Mr. Arif, Mr. Yash, Mr. Prakash, Mr. Tarakesh, and Ms. Rishika for helping me out in difficult times. I would also like to thank my batchmates for the healthy discussions and support during my thesis work. I am grateful to my friends who have willingly helped me with the best of their abilities and kindness.

Of course, no gratitude is complete without acknowledging my family members, who have ingrained in me virtues such as determination, self-respect, and patience. Throughout my journey, they have always been a source of inspiration and unwavering support. Above all, I owe it all to Almighty God for granting me the wisdom, health and strength to undertake this research task and enabling me to its completion.

Sanjay Adhikary

Contents

List of Tables vi

List of Figures viii

List of Abbreviations viii

Abstract ix

1 Introduction 1

2 Literature review 3
2.1 Assimilative Capacity . . . 3
2.2 Literature study on assimilative capacity . . . 3
2.2.1 Estimation of Assimilative Capacity for Gangtok City . . . 3
2.2.2 Assimilative capacity assessment of kochi region for the sitting of environment friendly new industries . . . 4
2.2.3 Assimilative Capacity assessment in a highly polluted industrial cluster in Manali . . . 5
2.2.4 SPM Assimilative Capacity estimation of Mundra Taluka . . . 5
2.2.5 Estimation of the Airshed's Assimilative Capacity in the Goa Iron Ore Mining Region . . . 6
2.2.6 Prediction of air pollution concentrations from numerous sources by using AERMOD and the WRF prognostic model 7
2.2.7 Estimation of Assimilative Capacity of air pollutants in the largest Petrochemical complex in Thailand . . . 7
2.3 Summary of literature review . . . 8

3 Research gap and objective of the study 9
3.1 Research gap . . . 9
3.2 Objective of the study . . . 9

4 Research methodology 10
4.1 Area of study . . . 10
4.2 Data . . . 12
4.2.1 Meteorological data . . . 12
4.2.2 Emission data . . . 12
4.3 Methodology . . . 12
4.3.1 Assimilative Capacity based on Ventilation Co-efficient approach . . . 13

4.3.2 Assimilative Capacity based on Pollution Potential 14

5 Results and discussion **18**
5.1 Meteorological observation 18
5.1.1 Wind speed 18
5.1.2 Temperature and humidity 21
5.1.3 Mixing height(MH) 23
5.1.4 Ventilation coefficient (VC) 26
5.2 Air quality modelling 32
5.2.1 Sulphur dioxide (SO_2) 34
5.2.2 Oxides of nitrogen (NO_x) 40
5.2.3 Comparison of model results 46
5.2.4 Assimilative capacity 47

6 Findings and conclusion **49**
6.1 Findings 49
6.1.1 Meteorological 49
6.1.2 Dispersion modelling 50
6.2 Conclusion 51
6.3 Future scope of the study 52

References **53**

List of Tables

4.1 Zones of India based on assimilative capacity (Source:NEERI 1990) 14
4.2 Co-ordinates of discrete receptors 15

5.1 Duration in consideration for different calculation 26
5.2 Seasonal variation of VC and corresponding assimilative capacity as per NEERI 26
5.3 Fixed input parameter for the point sources for both the pollutants. (Source:EIA report of refinery) 32
5.4 Variable stack input data for SO_2. (Source:EIA report of refinery) . 34
5.5 Variable stack input data for NO_x. (Source:EIA report of refinery) 40
5.6 Comparison of predicted GLC's at discrete receptors for SO_2 46
5.7 Comparison of predicted GLC's at discrete receptors for NO_x . . . 47

List of Figures

4.1 Image of study area . . . 11
4.2 Distribution of uniform and discrete receptors over the study domain 15
4.3 Typical working of AERMOD . . . 16
4.4 Dispersion modelling flow chart . . . 17

5.1 Seasonal variation of WR diagram . . . 19
5.2 Seasonal variation of WR diagram . . . 20
5.3 Monthly variation of temperature . . . 21
5.4 Seasonal variation of temperature and humidity . . . 22
5.5 Procedure of obtaining mixing height. Source: (Ashrafi *et al.* 2009) 23
5.6 Monthly variation of mixing height . . . 24
5.7 Seasonal variation of mixing height . . . 25
5.8 Variation of VC for winter . . . 27
5.9 Variation of VC for summer . . . 28
5.10 Variation of VC for monsoon . . . 29
5.11 Variation of VC for post monsoon . . . 30
5.12 Monthly variation of mixing height . . . 31
5.13 Position of the stacks . . . 33
5.14 Winter SO_2 variation of predicted concentration over the study domain . . . 35
5.15 Summer SO_2 variation of predicted concentration over the study domain . . . 36
5.16 Monsoon SO_2 variation of predicted concentration over the study domain . . . 37
5.17 Post monsoon SO_2 variation of predicted concentration over the study domain . . . 38
5.18 Annual SO_2 variation of predicted concentration over the study domain . . . 39
5.19 Winter NO_x variation of predicted concentration over the study domain . . . 41
5.20 Summer NO_x variation of predicted concentration over the study domain . . . 42
5.21 Monsoon NO_x variation of predicted concentration over the study domain . . . 43
5.22 Post monsoon NO_x variation of predicted concentration over the study domain . . . 44
5.23 Annual NO_x variation of predicted concentration over the study domain . . . 45
5.24 Seasonal variation of assimilative capacity . . . 48

List of Abbreviations

AC Assimilative Capacity

AMS American Meteorological Society

AQM Air Quality Model

CDU Crude Distilattion Unit

DALR Dry Adiabatic Lapse Rate

DCU Delayed coker unit

DG Diesel Generator

EIA Environmental Impact Assessment

ELR Environmental Lapse Rate

GLC Ground Level Concentrations

IMD India Meteorological Department

MH Mixng Height

NAAQS National Ambient Air Quality Standards

NEERI National Environment Engineering Research Institute

SDU Solvent de-oiling unit

SPM suspended particulate matter

USEPA United States Environmental Protection Agency

VC Ventilation Coefficient

VDU Vacuum distillation unit

VOC Volatile Organic Compounds

WR Wind Rose

WRF Weather Research and Forecasting

Abstract

In recent times we often talk about climate change and it has already started showing its devastating effect to the environment and the researchers have shown that both air pollution and climate change are the two sides of a coin. Being a developing nation or for a developed nation also it is not possible to shift overnight pollution free energy sources.Hence, it necessitates to establish a balance so that development work and environment safety goes hand by hand. So, to control the air pollution firstly we must aware about the pollution trends and their dispersion from our existing sources and one of the most suitable tool is to use dispersion modelling.

In this study we are focussing on the dispersion modelling of gaseous air pollutants majorly SO_2 and NO_x from the Digboi oil refinery. Firstly based on ventilation coefficient approach an estimate is being made about the range of pollution potential of that region.Then we have gone for dispersion modelling based on existing pollution load of that area from the Digboi oil refinery only and at last an estimate is also being made on the available assimilative potential of that region based on the max permissible load an area can bear with based upon NAAQS standard.

Based on ventilation coefficient approach it has been observed that the area has low assimilative potential for all the seasons and also evening value of ventilation coefficient always exceeds that to its morning value as the same thing can be observed from different studies also.

Model predicted result at existing load shows monsoon is the critical season for both the pollutants but for all receptors(uniform and discrete) predicted ground level concentrations are well below the permissible limits clearly indicating about the area's fair available assimilative potential.

Max available assimilative capacity value of 2702.6 kg/d for NO_x is observed during summer while max available assimilative capacity value of 3665 kg/d of SO_2 is observed during post monsoon. Also at the max pollution load at all the discrete receptors the predicted concentration of pollutants are well below the permissible limits.

Keywords: AERMOD,Mixing Height, Ventilation Coefficient, Assimilative Capacity, Pollution potential.

Chapter 1

Introduction

Air pollution is one of the most serious issues facing the earth today. Pollutants are constantly discharged into the atmosphere for a variety of causes, including the creation of new industrial states and the increasing number of motor vehicles combined with population increase, particularly in urban regions (Krishna *et al.* 2004). High population density and economic development characterized urban areas. As a result of the pollutant releases, the air quality in these locations is deteriorating. Whereas in the past, industrial activity and household heating were the primary causes of poor air quality, today, as a result of the rapid growth in mobility, road traffic is the primary source of modern urban air pollution (Sturm *et al.* 2000). As the number of air pollution incidents rises, the population becomes more vulnerable to asthma attacks, bronchitis, and even if it may be led to the premature death (Holloway et al. 2005). As a result of the public's increased knowledge of the negative consequences of air pollution, policy-making agencies have begun to integrate air pollution control methods as an intrinsic aspect of urban planning. As a result, these regulatory organizations test the ambient air quality on a regular basis. Regular monitoring of contaminants through dense air quality monitoring networks, on the other hand, is costly and not always practical (Krishna *et al.* 2005).

Alternatively, air quality modelling can offer valid information on air quality by filling up the gaps between monitoring stations (Zou et al., 2010). The air quality modelling is simple yet reliable, and it is quickly becoming an important part of all urban air quality management program. By taking into account atmospheric dispersion, chemical and physical processes, an air dispersion model aims to forecast the spatiotemporal fluctuation of outdoor contaminants (Holmes and Morawska, 2006). The AERMOD, for example, is a state-of-the-art regulatory dispersion model created by the American Meteorological Society (AMS) and the United States Environmental Protection Agency (USEPA) for a wide range of applications.

Air pollution has become a major source of concern, especially in mega-cities and metropolitan regions, where the situation is frightening and worsening day by day, necessitating careful planning to support future industrial development. Site selection is a must, especially for air polluting enterprises, with the goal of limiting negative environmental consequences based on environmental criteria. In order to site any air polluting enterprise, the region's assimilative capacity must be carefully examined and planned for, so that the receiving environment is not

harmed.

Refineries and petrochemical industries, like other industrial activities, can have an impact on air quality. They may emit pollutants such as VOC's (Volatile Organic Compounds), particulate matter, greenhouse gases, and contribute to ozone formation, primarily as a result of separation, conversion, treatment, combustion, storage, and unexpected leakages from process equipment, as well as auxiliary sources such as cooling towers, boilers, and sulphur recovery units. These emissions degrade air quality, as measured by Air Quality Indices, and air quality management is essential for resolving the issue and minimizing the damage.

Digboi industrial area was continuously under non attainment city list upto 2018 as per CEPI action plan and the area is filled up with oil wells,coal mines, tea manufacturers and the Asia's oldest oil refinery. Majorly they were polluting the environment but their continuously improvement of emissions into the environment shows the they were able to came out of this non attainment city list. But the major contributor was the oil refinery and also in recent times they are expanding their production from 0.65MT to 1MT per year and hence in my present study i am focussing on the dispersion modelling of gaseous air pollutants from the oil refinery as to evaluate the status of pollution concentration over the region due to that refinery.

Hence in the present study focuses on estimation of assimilative capacity through ventilation coefficient and air quality modelling approach in the Digboi industrial area .

Chapter 2

Literature review

Literature review

This chapter overviews on Assimilative Capacity and the approaches by which we estimate the assimilative capacity i.e. ventilation coefficient and pollution potential and also different parameters influence on those methods. Then we will see some case studies based on those methods applied on different region in different times to assess the atmosphere's assimilative potential. Finally, we will make a summary of the chapter as well.

2.1 Assimilative Capacity

It is the maximum air pollution load an area can take without violating the permissible concentrations of pollutants or else we can say without causing any harm to the ambient environment and its receptors. Basically, it is found out by using two approaches. The first one is through ventilation coefficient (mixing height $\times$ mean wind speed) and the assimilative directly proportional to the this. Secondly through any USEPA approved dispersion modelling tools like AERMOD, ISCST, CALINE-3 etc..These tools uses different data such as meteorological, emission and terrain and predict the maximum ground level concentrations to receptors which we validate with observed data at the nearby station through any statistical tools like RSME and MAE etc..

Then the difference between max permissible pollutant concentrations and present pollution concentrations predicted through the modelling tool will give us the assimilative capacity of atmosphere of that region. from there we can easily find out the allowable air pollution load of that region.

Several literature studies are shown in the next section.

2.2 Literature study on assimilative capacity

2.2.1 Estimation of Assimilative Capacity for Gangtok City

The assimilative capacity of the atmosphere was evaluated in two ways. The ventilation coefficient was calculated using the product of mixing height and average wind speed. The diurnal variation of ventilation coefficient values indicates that

winter is the season with the worst weather. However, this method is insufficient to make a conclusion. The highest value of ventilation coefficient was 3135 m^2/s in April 2004 and 983 m^2/s in December 2004.

The second method involves estimating the atmosphere's assimilative capacity by using dispersion models to calculate the concentration contribution from line sources to the ambient air. Because auto mobile traffic is the primary source of pollution, the pollution potential is calculated using the Caline-3 and IIT Line Source model.

In December, the concentration of SO_2 predicted by IITLS ranges from to 33 mg/m^3, while in April, it is between 4 and mg/m^3. Similarly, in December, SPM ranges from 45 to 168 mg/m^3 , but in April, it falls to a range of 40–161 mg/m^3. In December, NO_x levels vary from 4 to 29 mg/m^3, while in April, NO_x levels range from 3 to 27 mg/m^3. In all seasons, lower concentration values occur in the afternoon hours, whereas greater concentration values occur in the early morning and evening.

The study region has a high contamination potential, meaning a low assimilative capability. Assimilation capacity is higher in the summer than in the winter. In both seasons, the IITLS model outperforms the Caline-3 model for predicting pollutant concentrations at receptors (P. Goyal *et al.* 2006).

2.2.2 Assimilative capacity assessment of kochi region for the sitting of environment friendly new industries

The assimilative capacity of an air volume of 24 km× 24 km× 100 m was calculated in this investigation. Further, a critical season was chosen, that is, the time of year when pollution dispersion is the least. With respect to suspended particulate matter (SPM) and sulphur dioxide, the likely occurrence of maximum pollutant concentration at each receptor in the research region was investigated for different emission loads (current and predicted future) (SO_2). The region's assimilative capacity was defined as the discharged emission load at which the maximum permissible concentration was reached under the given critical circumstances. The region's assimilative capability was designated as one of the crucial conditions.

The ISCST model, in general, requires information about the source location, source emission characteristics, current weather conditions, and receptor locations. The following parameters are required as inputs: source location (x, y, z), m; stack height (m); stack diameter (m); exit velocity (m/s); temperature of stack gas (K); and pollutant emission rate (g/s) Hourly meteorological data is also required, including wind direction (degrees), wind speed (m/s), ambient air temperature (K), stability class (1–6), and mixing height (m).

The model was solely used to estimate the spatial distribution of pollutants (SPM and SO_2) from point sources in this investigation.

The initial model runs' pollutant concentration levels at various emission loads were assessed. If the predicted maximum concentration in the study area was found to be less than the maximum permissible concentration, the emission load from all stacks in the study area was increased uniformly until the maximum allowable concentration levels were reached. The assimilative capacity of the study region under the above stated parameters was calculated using the total emission

load obtained in the final run.

From the model run it has observed that, the cumulative maximum GLC was observed to occur at a distance of 4.1 km in the SW direction (with respect to the meteorological station) during the winter, and at a distance of 22.7 km in the NW direction during the post-monsoon.For respective SPM emission loads of 43,200, 17,280, and 8640 kg/d, the cumulative maximum ground level concentration(GLC) was expected to be 273, 109, and 55 mg/m^3 during the winter. The same were 390, 156, and 78 mg/m^3 during the post-monsoon period, respectively.

The maximum GLCs of SPM and SO_2 predicted under various emission and meteorological scenarios show that the post-monsoon (October) season has less dilution/ dispersion potential than the winter (January); thus, the post monsoon season's emission load is found to be critical (S. Goyal & Rao 2007).

2.2.3 Assimilative Capacity assessment in a highly polluted industrial cluster in Manali

In this study, for assessing assimilative capacity, the American Meteorological Society/US Environmental Protection Agency Regulatory Model (AERMOD) was employed to quantify the assimilative capacity (i.e., the maximum emission an area can take without violating permissible pollutant standards) of a major industrial cluster (Manali, India) and to evaluate the effectiveness of the region's adopted air pollution control measures. To determine the best and worst circumstances for industrial operations, seasonal analyses of assimilative capacity were conducted according to critical, high, medium, and low pollution levels.

Sulphur dioxide(SO_2, nitrogen dioxide (NO_2), and particulate matter (PM_{10}) emissions in the Manali industrial cluster were measured using a bottom-up technique with a fine spatial resolution of 500 x 500 m^2. Industries of Manali area emitted 22.8 ton of SO_2, 7.8 ton of NO_2, and 7.1 ton of PM_{10} per day, according to the findings. The SO_2, NO_2, and PM_{10} assimilative capabilities were reported to be 16.05, 17.36, and 19.78 ton/day, respectively. Current SO_2 emissions were found to be 6.7 ton/day more than the predicted permissible load, whereas PM_{10} and NO_2 levels were within safe limits. Seasonal analysis of assimilative capacity revealed that the post-monsoon season had the lowest load-carrying capacity, followed by winter, summer, and monsoon seasons, with allowable SO_2 emissions being found to be 35 percent and 26 percent lower, respectively, when compared to the monsoon season (Panda & Nagendra 2017).

However, a similar study was carried out in this region by Mnaju et al.(2002) through ISCST model and according to their observations, the assimilative capacity of the Manali region was best in the summer and lowest in the winter (Manju *et al.* 2002).

2.2.4 SPM Assimilative Capacity estimation of Mundra Taluka

For the estimation of Assimilative Capacity of the area two separate model runs were performed for point and area sources to find out the predicted SPM ground level concentrations (GLCs).

All industrial emissions were considered as point sources, whereas in area sources domestic and vehicular emissions were considered.

Model prediction shows that the industrial sector accounts for 91% of the overall air pollution load in the study region, followed by the domestic sector (2%), and the vehicle sector (2%).

The prediction of pollutant concentration was done for a receptor grid size of $1000m \times 1000m$ and the area was further subdivided into the smaller grids of size $400m \times 400m$. SPM has a total predicted maximum average 24 hourly concentration of 55 $\mu g/m^3$. ISCST modelling was used to determine the current average ambient concentration in the region. The available increment is 145 $\mu g/m^3$ because the 24- hourly averaged NAAQS for SPM is 200 $\mu g/m^3$ and the maximum anticipated GLC from ISCST modelling for SPM is 55 $\mu g/m^3$. The region's supporting load was assessed to be 19318.35 Kg/d for this available increment (Shaikh 2010).

2.2.5 Estimation of the Airshed's Assimilative Capacity in the Goa Iron Ore Mining Region

The study of Ghose MK et al. (2000,2001) and Huertas J et al. (2012) shows that particulate matter is the most common and most released air pollutant in mines, with a chemical composition comparable to that of re-suspended particles in the surrounding area (M. K. Ghose & Majee 2000) (M. Ghose & Majee 2001) (Huertas *et al.* 2012). Further the study of Chaulya SK et al. (2001) and several others show that, in the mining industry, material transportation is the primary source of Total Suspended Particulate (TSP) matter generation. TSP and PM10 concentrations are highest in a mining region, and concentrations gradually decrease as distance increases due to particle transportation, deposition, and dispersion. Particulate matter dispersion is influenced by an area's yearly prevalent wind direction (Singh & Perwez 2015).

Here Gurdeep Singh et al. (2015) for this study selects 14 receptors sensitive area with due regard for the region's sensitivity as well as population exposure. Then the assimilation potential of the atmosphere was calculated in terms of potential pollution load using the American Meteorological Society EPA Regulatory Model (AERMOD) dispersion model to compare expected PM10 concentrations at those 14 discrete receptor locations.

Hourly surface meteorological data from IMD, Pune, and upper air meteorological data from AERMOD upper air estimator were used in this study. Boundary layer parameters such the Monin-Obukhov length, convective velocity scale, temperature scale, mixing height, and surface heat flux were calculated using these meteorological data for AERMET. The second module, AERMAP, is utilized to process topography data in conjunction with a receptor and emission source layout for usage in the AERMOD input files.

PM_{10} concentration was found to be well below the permitted limit ($60\mu g/m^3$) at all of the receptors studied. The maximum assimilative capacity was 53 $\mu g/m^3$, while the lowest was 23 $\mu g/m^3$. According to the above calculations, the area under examination has a reasonable and acceptable assimilative capacity and can dissipate an adequate pollution load.

2.2.6 Prediction of air pollution concentrations from numerous sources by using AERMOD and the WRF prognostic model

This research aims to collect SO_2,NO_2, and PM_{10}dispersion data from 45 major industrial sources in Pasir Gudang, Iskandar region, Johor, Malaysia. Malaysia's concerns about poor air quality may soon surface as a result of the rapid economic growth in the Iskandar region in Johor's south. The spatial variations of SO2, NO2, and PM10 emitted from multiple industrial sources in Pasir Gudang industrial area, Johor, Malaysia, were predicted using an air dispersion model developed by the American Meteorological Society/Environmental Protection Regulatory Model (AERMOD) and Weather Research and Forecasting (WRF) . For a two-week period, the WRF model was used to replicate hourly surface and upper air meteorological data. Temperature, wind speed, and wind direction, among other WRF output factors, were statistically assessed in the study. The model was evaluated using statistical criteria such as RMSE (Root Mean Square Error) and MAE (Mean Absolute Error).

The maximum ground level concentrations of SO_2,NO_2, and PM_{10} simulated by AERMOD-WRF in the industrial region were 36.2, 59.8, and 5.4 mg/m3, respectively, according to the data. The Quantile-Quantile (Q-Q) plots used to evaluate AERMOD revealed that the majority of the expected and observed pair points are near to the one-to-one line, indicating that the predicted and observed concentrations in the study are in good agreement. The outcomes of this study can help local government officials and policymakers manage urban air quality more effectively (Afzali *et al.* 2017).

2.2.7 Estimation of Assimilative Capacity of air pollutants in the largest Petrochemical complex in Thailand

The assimilative capacity in terms of Sulphur dioxide(SO_2) and nitrogen oxides (NO_x) were assessed in this study.

The ground level concentrations and spatial distributions of these contaminants within the research region were computed using the AERMOD dispersion model. The AERMOD modelling was utilized in this study to determine the maximum ground-level pollutant concentrations using a thorough Cartesian receptor grid extending to 5 km from the center of the emission source. The consistent spacing of 500 meters is used in the Cartesian receptors grid (14.5 km x 14.5 km domain). In the modelling domain, there were eight distinct receptors.

Emission input consisted of 419 stacks with total SO_2 and NO_x emissions of 2,071.82 and 2,163.66 g/s, respectively. The assimilative capacity of these pollutants was evaluated using maximum hourly average and yearly concentrations projected at 8 ambient air quality monitoring stations in the study domain.SO_2 and NO_x emissions have been modified to ensure that anticipated concentrations do not exceed ambient air quality regulations.

According to the findings,NO_x emissions must be decreased by at least 40% from their baseline value in order to meet both hourly and annual limits. In the case ofSO_2, emissions might be raised by up to 130 percent over existing levels. The findings of this analysis show that the current regulation for limiting individ-

ual emission sources through the use of emission standards is insufficient. Analysis of air pollution as an area-based approach, taking into account all emission sources, can aid in improved pollution planning and management (Thawonkaew *et al.* 2016).

2.3 Summary of literature review

From the above discussion in summary, we can say that assimilative potential (i.e. the atmosphere's ability to receive and disperse pollutant discharge without exceeding acceptable limits) of the atmosphere can be found out by two approaches. First is ventilation coefficient (i.e. average wind speed × mixing height) which gives an overall idea about the load carrying potential of the atmosphere for ex low, medium or high and the assimilative capacity is directly proportional to it.

Second is through pollutant potential and assimilative capacity is inversely proportional to it.in this approach assimilative capacity of the atmosphere can be found out by subtracting present pollutant potential from the max permissible pollutant potential (as per NAAQS) of that region. To find out present potential we use any of US- EPA approved dispersion modelling software like AERMOD, ISCST-3 etc..

In which we divide the study area in smaller grids and on every grid's meteorological, pollutant concentrations, terrain characteristics are used as input to the model and then we run the model to find out the actual GLC of the pollutants. For the entire area they found out the cumulative average of the pollutant concentrations. Statistical tools like RSME (root means square) sometimes Q-Q plots are used for the validation of the model. Most the times the model gives the satisfactory performance and the corresponding results were disclosed to the concerned authorities and based on that necessary action was taken.

Chapter 3

Research gap and objective of the study

3.1 Research gap

Based on different literature survey on Assimilative Capacity determination as per modelling and simulation approach shows several drawbacks. In which the most important is the elimination of background pollution and emissions from automobile traffic, which accounts for a considerable portion of particulate pollutants. Due to that most of the time the variation between predicted and observed concentration of pollutants were noticed. Sometimes the weather station at which observed concentrations were measured was outside the study area and in between background and vehicular sources make a significant impact which led to the variations of observed and predicted pollutant concentrations from the model. It is also observed that the meteorological parameters provided to AQMs may have uncertainties that effect model simulations negatively.

3.2 Objective of the study

- Dispersion modelling of gaseous air pollutants from the Digboi oil refinery.
- To find out the impact of gaseous air pollutants at discrete receptors.
- Assessment of assimilative capacity of the atmosphere of Digboi industrial area.
- Estimation of the max air pollution load the area can support without violating the NAAQS standard.
- To find out the critical period i.e the time period in which the atmosphere has the least pollution carrying potential due several meteorological parameters.
- To find out the seasonal variation of assimilative capacity of the ambient atmosphere of the area.

Chapter 4

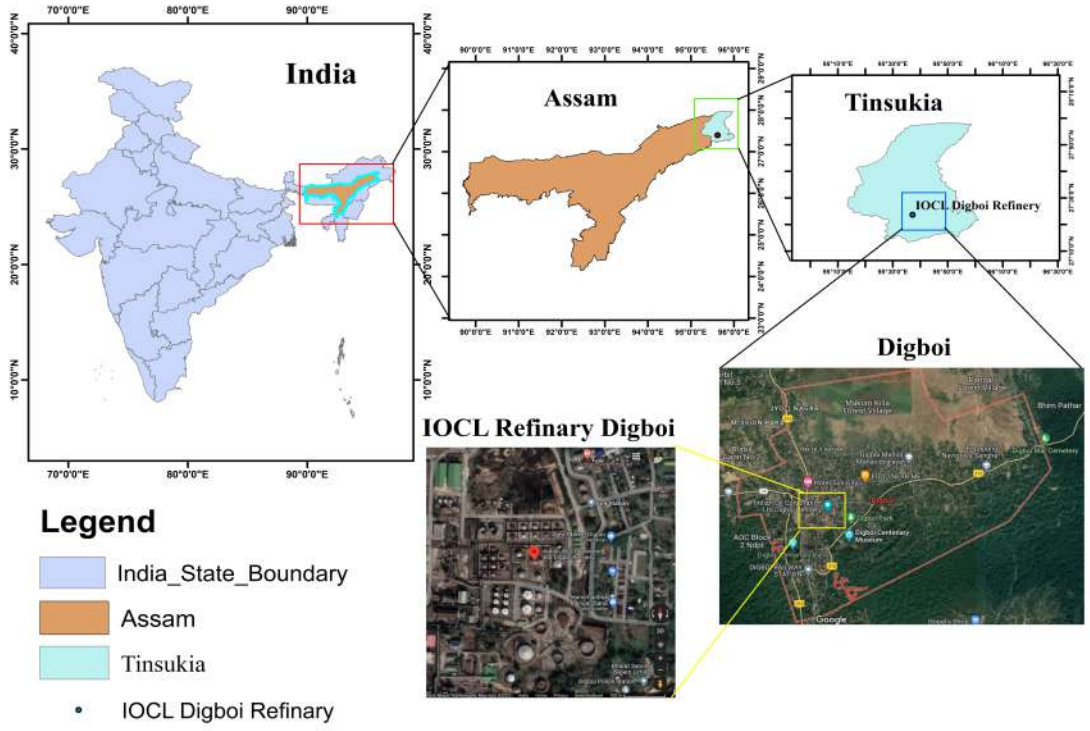
Research methodology

4.1 Area of study

Digboi industrial area is chosen for this study which is in a town and in Tinsukia district of Assam in the north-eastern part of the state of Assam, India and its co-ordinates read 27.38°N and 95.63°E. It spreads over 14.8 Km^2 and more than 22000 (as per 2011 census) peoples are living there. In major industries Digboi refinery is there which is also known as the birthplace of the petroleum refining industry in India. A picture of the Digboi area and Digboi refinery are shown in, 4.1a and 4.1b.

The major problem with the area is the emissions from the oil refinery. The refinery consumes fossil fuels for its operational purpose due to which greenhouse gases like methane, carbon dioxide, sulfur dioxide and nitrous oxide are continuously emitting into the environment. The refinery releases significant amounts of VOC's also and this led to the generation of Particulate matters in the atmosphere. More over domestic burning and vehicular emissions also playing a significant role to the pollutant concentration contribution.

Now our focus will be on to assess the atmospheres assimilative potential in the present scenario and draw a conclusion about the critical period of pollution over the region due to these combined emissions.



(a) Map of study area



(b) Satellite image of digboi refinery

Figure 4.1: Image of study area

4.2 Data

This section describes the emissions data, including emission rate and stack characteristics, meteorological data, and observed air quality data for the year 2019.

4.2.1 Meteorological data

The meteorological data for the entire 365 days of 2019 are taken in this present study. The 3-hour wind speed, wind direction, temperature, cloud cover, and solar insulation are among the meteorological data. These data are collected by the India Meteorological Department in Dibrugarh and monitored at the Dibrugarh airport. The airport is located close to and within the study area.

Using the Holzworth (Holzworth 1967) technique, the mixing height (The mixing height is the distance above the surface at which relatively rigorous vertical mixing takes place)is determined. Thus, 3-h (0000, 0300, 0600, 0900, 1200, 1500, 1800, and 2100 h) meteorological data on wind speed, wind direction, temperature, air stability, and mixing height were used in the current investigation. Based on the frequency distribution of wind direction as well as wind speed, a reasonable approximation of pollution dispersion in the atmosphere can be made (Manju *et al.* 2002). Wind rose diagrams are plotted for all seasons namely - summer, winter, monsoon and post monsoon at Digboi industrial area are shown in the fig 5.1 and fig 5.2.

The prevailing wind direction for all the seasons is NE and E but during monsoon there is a significant amount of wind blows from SW directions also. Calm condition for summer season reads 6.52% and that for winter monsoon and post monsoon are 23.31%, 8.71% and 34.38% respectively. Hence is shows during winter and post monsoon more calm condition prevails.

4.2.2 Emission data

Only one major industry and that is the Asia's oldest Digboi oil refinery is there as shown in the 4.1b and it has 6 major stacks namely CDU (Crude Distilattion Unit) , VDU (Vacuum distillation unit) , DCU (Delayed coker unit) , SDU (Solvent deoiling unit) , CRU (Catalytic reforming unit, 2 units HDT and OBSG), HDT and HGU along two stacks for the each of diesel generator set (DG set 1,2) . Thus, a total number of 8 stacks of the refinery is considered as elevated point sources in the present study. The location (x,y) of each stack on a Cartesian grid is taken into account. The source strengths or emission rates of two gaseous pollutants, namely, SO2 and NOx are considered. In the present study, I have taken into account the stack characteristics such as stack height, stack internal diameter, stack exit gas velocity and stack exit temperature of each stack.

4.3 Methodology

Assimilative Capacity

The ability of the environment to carry contaminants without causing harm to the environment or users of its resources is referred to as assimilative capacity.

According to Manju et al. (2002), the atmosphere's assimilative capacity specifies the maximum pollution load that can be emitted into the atmosphere without breaching the planning region's best-designed use of air resources (Manju *et al.* 2002). There are two methods for determining the atmosphere's assimilative capacity. The first is based on the ventilation coefficient, which can be determined using meteorological information, and the second is based on the pollution potential due to the existence of numerous air pollution sources in the research area. The pollution potential describes the atmosphere's ability to dilute pollutants and the subsequent consequences on air quality (Krishna *et al.* 2004).

4.3.1 Assimilative Capacity based on Ventilation Co-efficient approach

Assimilative Capacity is directly proportional to the Ventilation Co-efficient(VC) which the multiplication of mean wind Speed (m/s) and mixing height(m). It has been observed that VC is the ability to reduce air pollution in an urban or rural environment. More clean air is produced when the VC is higher. As a result, this critical parameter can be used as a metric in urban air pollution strategies, programs, and simulations (Ashrafi *et al.* 2009).

The ventilation coefficient is calculated for two times morning (5:30AM) and evening (5:30PM) as we have calculated mixing height for two times as per Holzworth method (Holzworth 1967) i.e. morning (5:30AM) and evening(5:30PM) and ventilation coefficient is the product of mixing height and average wind speed at 10m from the ground level.

Mean wind speed

Average hourly wind speed of that area at the level of mixing height. To find out the major wind directions of that area we will prepare a wind wise plot based on the wind speed and its directions data.

Mixing height

The mixing height (MH) , which reflects the depth through which pollutants are vigorously mixed, is a significant measure in calculating air pollution concentrations near the ground (Berman *et al.* 1999). ABL has a direct relationship with Maximum Mixing Depth, which is the height at which a pollutant can travel vertically up to and partially into the inversion layer.

According to the United State National Meteorological Centre the assimilative capacity of an area is considered to be

- High: If the observed VC of that area is $> 6000 m^2/s$.
- Medium : If the observed VC of that area is between 2000 to 6000 m^2/s.
- Low : If the observed VC of that area is $< 2000 m^2/s$.

The same is mentioned in the study conducted by B. Majumder for the calculation assimilative capacity based on ventilation coefficient approach (Prakash *et al.* 2017).

The National Environment Engineering Research Institute (NEERI) in 1990 has divided India into nine sorts of zones based on the ventilation coefficient (a parameter that represents atmospheric assimilative capacity). At intervals of 2000 m^2/s, ventilation coefficients were categorized in the range of less than 2000 to more than 18,000 m^2/s which is shown in the table 4.1.

Table 4.1: Zones of India based on assimilative capacity (Source:NEERI 1990)

Assimilative capacity	Ventilation coefficient (m^2/s)	Zones	Pollution potential
Low	< 2000	1	High
Low	2000 - 4000	2	High
Low	4000 - < 6000	3	High
Medium	6000 - 8000	4	Medium
Medium	8000 - 10000	5	Medium
Medium	10000 - < 12000	6	Medium
High	12000 - 14000	7	Low
High	14000 - 16000	8	Low
High	16000 - < 18000	9	Low

4.3.2 Assimilative Capacity based on Pollution Potential

Assimilative Capacity is inversely proportional to the pollutant concentrations of the atmosphere and the gap between the allowable and existent pollutant concentration levels for the study region was used to calculate assimilative capacity i.e.

$$\text{Available assimilation potential} = \text{Permissible standard–Existing pollutant load.} \tag{4.1}$$

Existing pollutant concentration can be found out by using any of the US EPA approved model such as ISCST -3, AERMOD.

AERMOD is a steady-state Gaussian dispersion model that can be used to estimate the concentrations of various ambient air pollutants emitted by various sources (Area, line and point) and equation it uses to find out the pollutant concentration is given below-

$$C(x,y,z) = [\frac{Q}{2\pi u \sigma_y \sigma_z}] \times [e^{-\frac{y^2}{2\sigma_y^2}}] \times [e^{-\frac{(z-H)^2}{2\sigma_z^2}} + e^{-\frac{(z+H)^2}{2\sigma_z^2}}] \tag{4.2}$$

where C is the concentration of pollutant ($\mu g/m^3$), x, y, z the receptor location on the cartesian coordinate system, Q the emission rate of pollutant (g/s), u the wind velocity at 10 m height(m/s), σ_z and σ_y are vertical dispersion coefficient(m) and the crosswind and H the effective stack height(m).

For the dispersion modelling as per the guidelines of CPCB to show the impact of air pollutants released from the oil refinery a radius of min 10 Km area must be considered (CPCB 2009).



Figure 4.2: Distribution of uniform and discrete receptors over the study domain

Here a study domain of $20km \times 20km$ is considered keeping the stacks of the refinery at the center of the study domain. The entire study domain is divided into 1600 uniform cartesian receptors with a grid spacing of $513m \times 513m$ both in X and Y direction along with four discrete receptors are considered in my study area and the details of the discrete receptors are shown in the table 4.2.

Table 4.2: Co-ordinates of discrete receptors

Serial no	Receptors name	UTM Easting	UTM Northing
1	Digboi railway station	758846.76	3031383.61
2	Digboi college	757685.26	3032728.62
3	Digboi high school	757458.74	3032912.37
4	Digboi civil hospital	759278.38	3032978.3

The dispersion model simulation is done mainly for two gaseous air pollutants namely Sulphur dioxide and oxides of nitrogen as per data available based on the environmental impact assessment(EIA) report of the refinery on which mainly emission rate of pollutants and discharge rate from the different stacks are available. The following fig 4.2 shows the distribution of uniform cartesian receptors over the study domain along with discrete receptors over the study region.

The typical working of AERMOD is shown in the following fig 4.3.

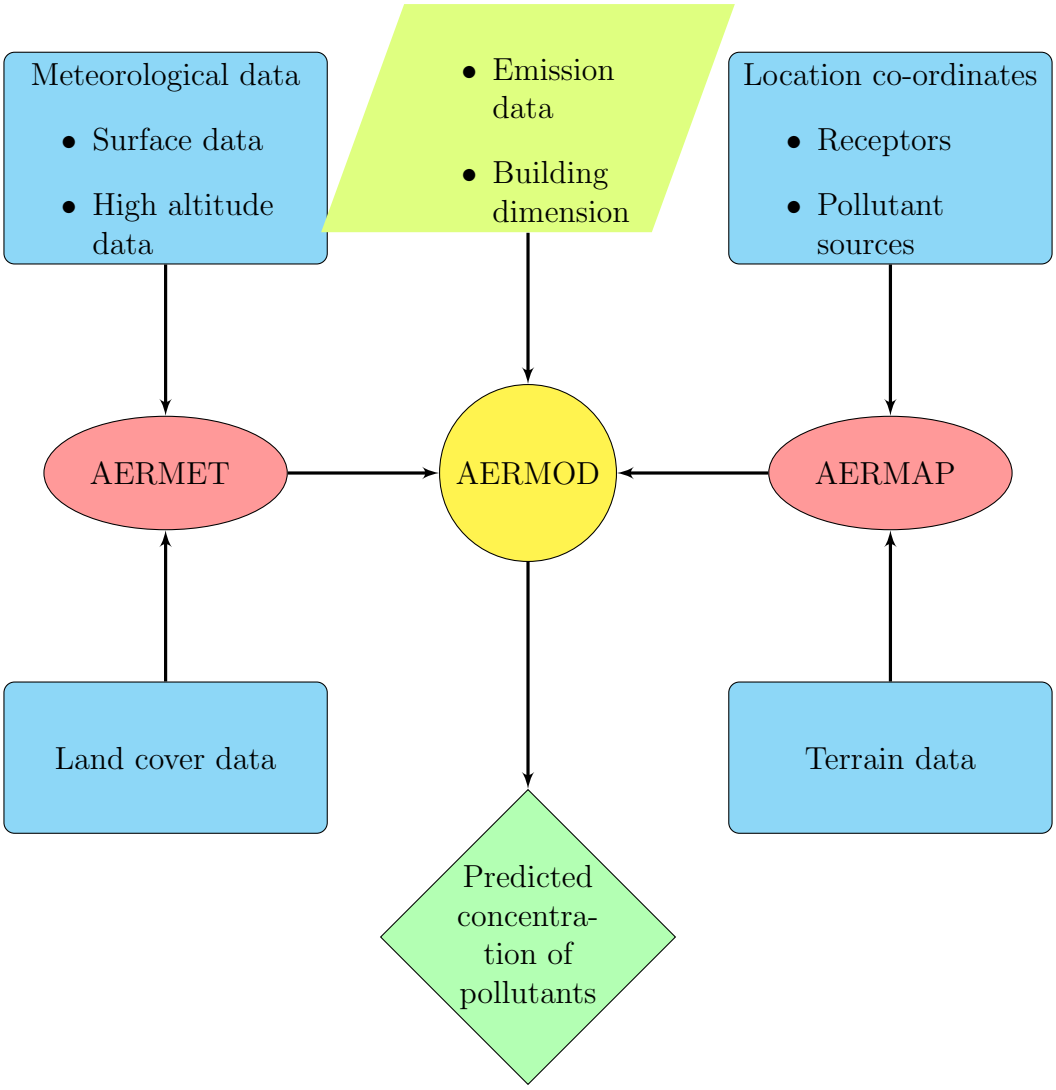


Figure 4.3: Typical working of AERMOD

Dispersion modelling flow chart for my current study is shown in the fig 4.4.

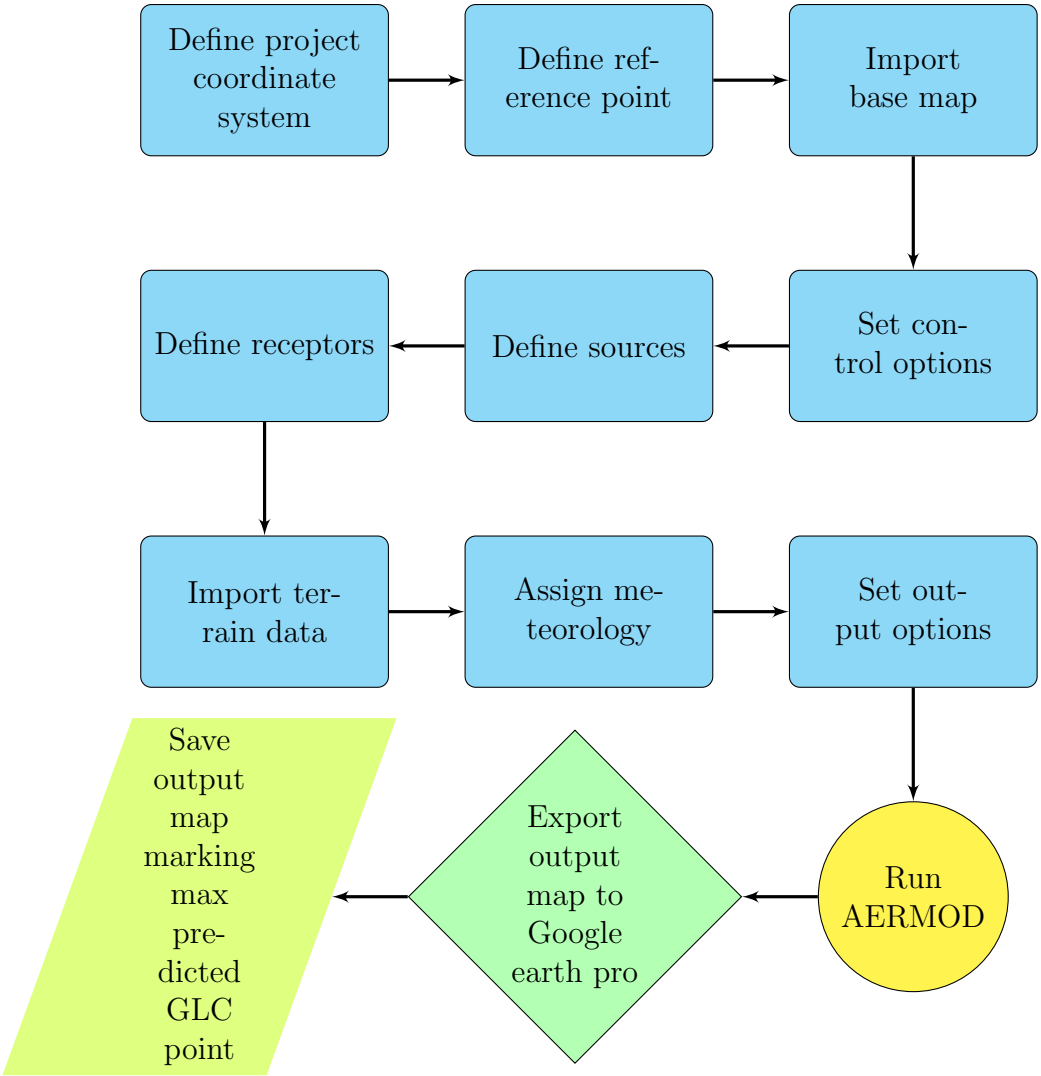


Figure 4.4: Dispersion modelling flow chart

Chapter 5

Results and discussion

5.1 Meteorological observation

The meteorological parameters such as wind speed, wind direction, temperature, humidity, mixing height etc.. and their corresponding seasonal and annual variations are discussed in this section.

5.1.1 Wind speed

Wind rose diagram (WR) plotting for all the four seasons has done and is shown below which shows the max wind speed of 11.54 m/s was observed on 27th march between 11 to 12 pm for the entire year.

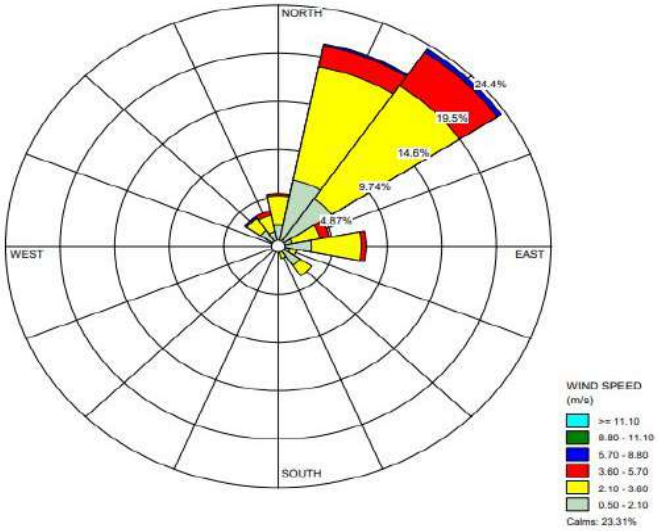
Max wind speed of 7.22 m/s for winter season fig 5.1a was observed on second February 2019 between midnight 0-1 am and average speed for the winter season is 1.93 m/s while the calm condition for the entire season is 23.3%.

Wind rose plot of summer shows fig 5.1b a calm condition of 6.52% prevails throughout the season while the max wind speed of 11.54 m/s on 27th march between 11 to 12 pm is recorded which is max among the all four seasons while the average wind of 2.53 m/s was observed for the entire season which is also max for the all four seasons.

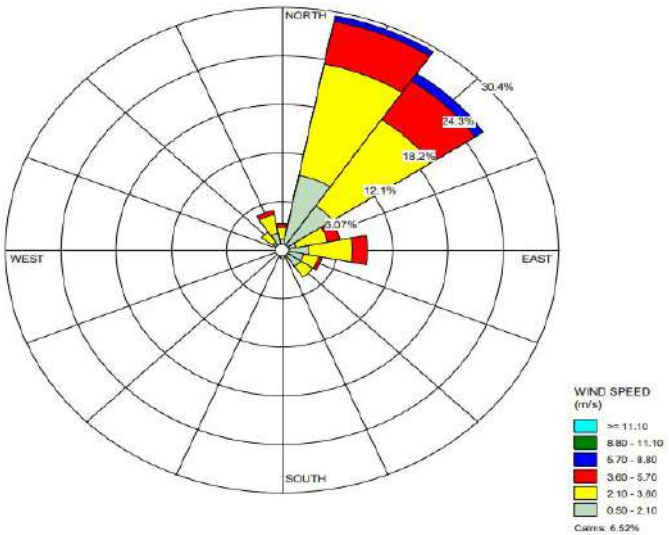
Wind rose plot of monsoon season shows fig 5.2a a calm condition of 8.71% was prevailed during the entire season while the max wind speed of 7.78 m/s was observed on 27th July between 3 to 4 PM and the average wind speed for the entire season was 2.5 m/s.

Wind rose plot of post monsoon season shows fig 5.2b that a max calm

condition of 34.38% was prevailed for the entire season as well as for the year 2019 also, a max wind speed of 6.11 m/s was observed



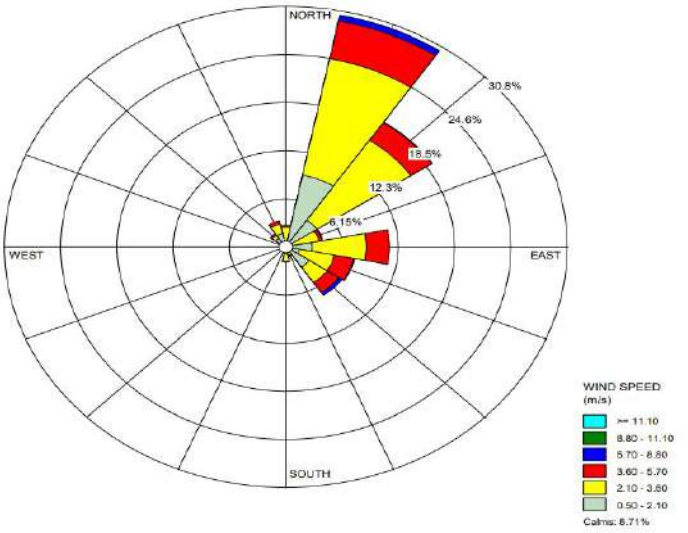
(a) Winter



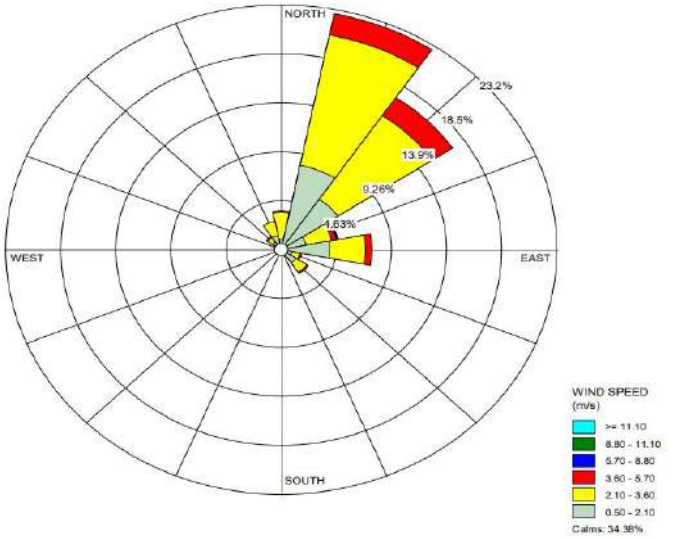
(b) Summer

Figure 5.1: Seasonal variation of WR diagram

on 10th November which is least for the all four season and an average wind speed of 1.56 m/s was maintained during the entire season which is also least among the all four seasons.



(a) Monsoon



(b) Post monsoon

Figure 5.2: Seasonal variation of WR diagram

5.1.2 Temperature and humidity

Temperature

Monthly and seasonal variation of maximum, minimum and average temperature is shown in the fig 5.3 and fig 5.4a for the year 2019. Monthly variation of temperature shows max temperature was observed on August which reads 37.5°C while minimum temperature was observed on December which reads 9.86°C. Monthly average variation of temperature shows that max average temperature of 29.71°C prevails during August and minimum average temperature of 16.66°C prevails during January.

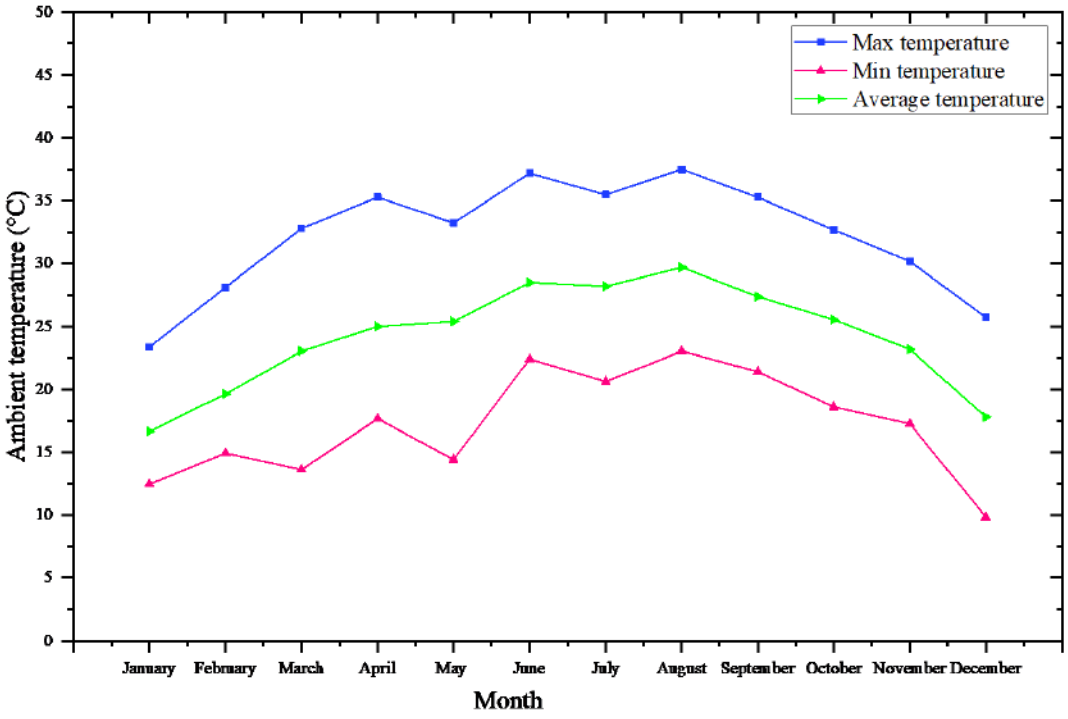


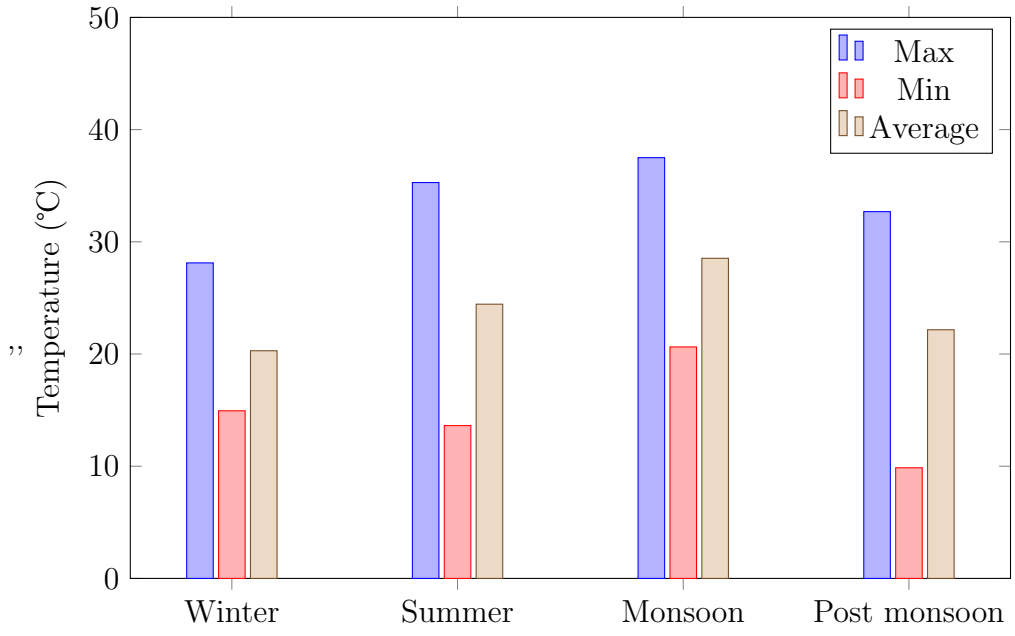
Figure 5.3: Monthly variation of temperature

Seasonal variation of temperature as shown in the fig 5.4a shows monsoon is the hottest season as both max (37.5°C) and average max (28.53°C) were observed during that period while minimum temperature of 9.86°C was observed on post monsoon but seasonal average minimum temperature of 20.29°C prevails during winter. Similar studies carried out by (S. Goyal & Rao 2007) for Kochi regions shows that Mean minimum and maximum temperature were recorded to be 22.3 and 32.41°C during winter and 23.3 and 29.71 °C during post-monsoon.

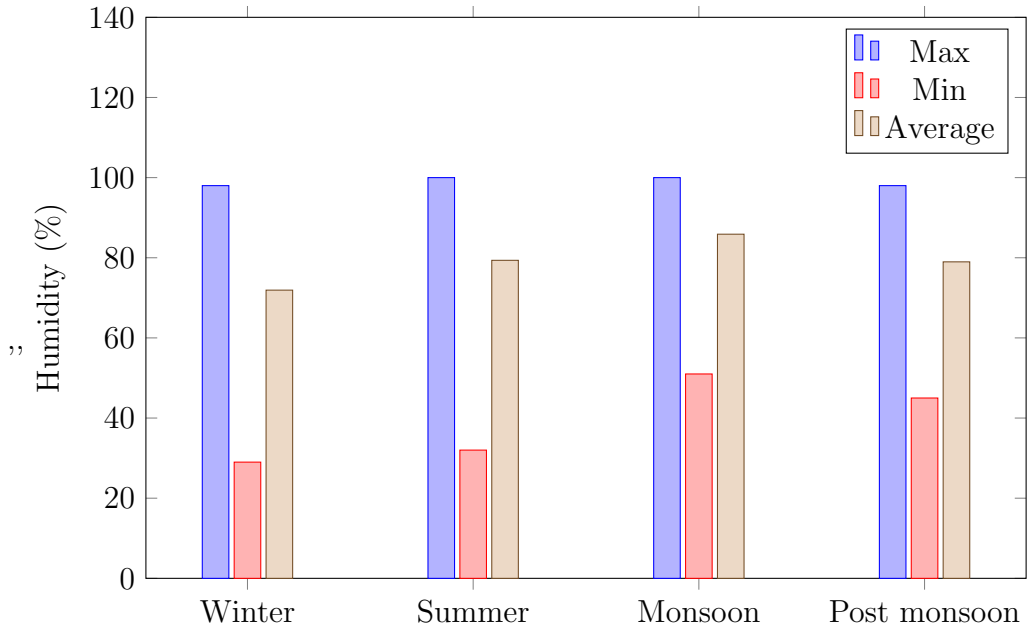
Humidity

Seasonal variation of relative humidity as shown in the fig 5.4b shows a max humidity of 100% prevails during both summer and post monsoon but a minimum humidity of 29% prevails during winter. Average maximum humidity of 85.88%

occurs during monsoon while average minimum humidity of 71.92% occurs during winter.



(a) Temperature



(b) Humidity

Figure 5.4: Seasonal variation of temperature and humidity

5.1.3 Mixing height(MH)

Mixing height is calculated as shown in the fig 5.5 as the height on which difference between environmental lapse rate (ELR) and dry adiabatic lapse rate (DALR) is zero, for ELR we have taken the hourly temperature variation data w.r.t height and DALR is fixed -10° C per kilometer of rise.

For the calculation of DALR temp variation with height we have taken min temperature (between 2 AM to 6 AM) plus 5° C for morning mixing height calculation and max temperature (between 12 PM to 4 PM) for evening mixing height calculation (considering urban heat island effect) and vary that -10° C per km above the ground surface (Holzworth 1967).

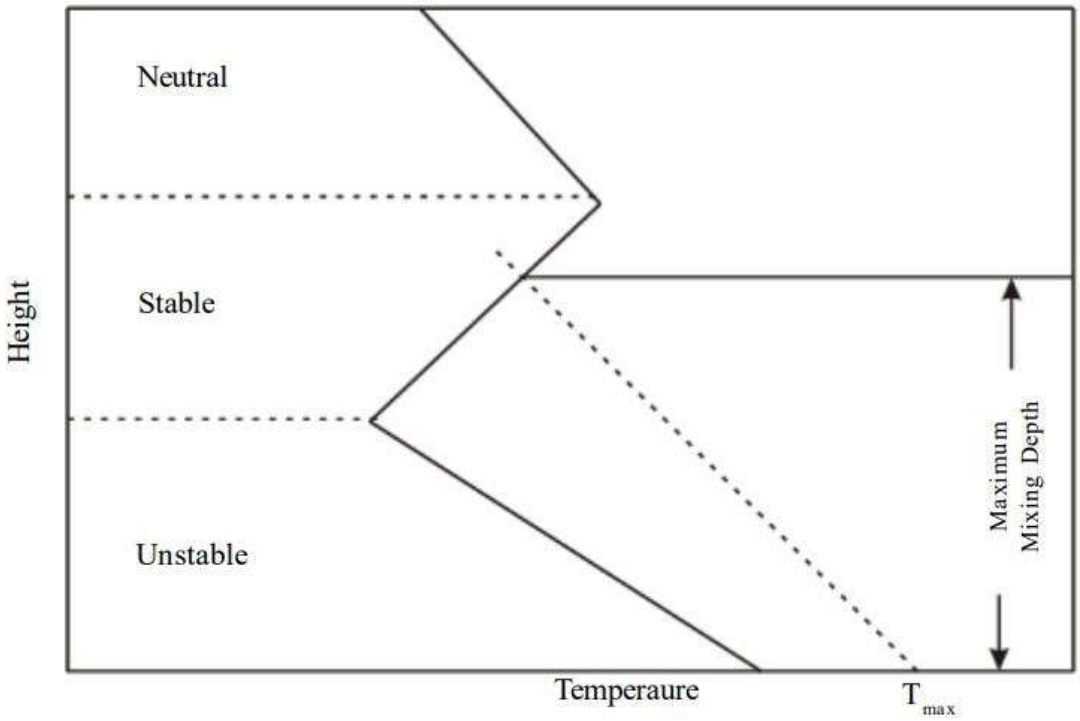


Figure 5.5: Procedure of obtaining mixing height. Source: (Ashrafi *et al.* 2009)

Monthly variation

Monthly variation of morning and evening average mixing height is shown in the fig 5.6 shows that evening hours have more mixing depths as compared to morning hours throughout all the months (January to November). For morning hours max average mixing depth value of 698.7 meter was observed during June and min average mixing depth value of 161 meter was observed during January.For evening hours max average mixing depth value of 1212 meter was observed during march and minimum average mixing depth value of 341 meter was observed during January. Similar studies carried out by (Holzworth 1967) for Los Angeles, New York and Pittsburg Columbia also shows that average mixing depth for evening hours are significantly more than that of morning hours.

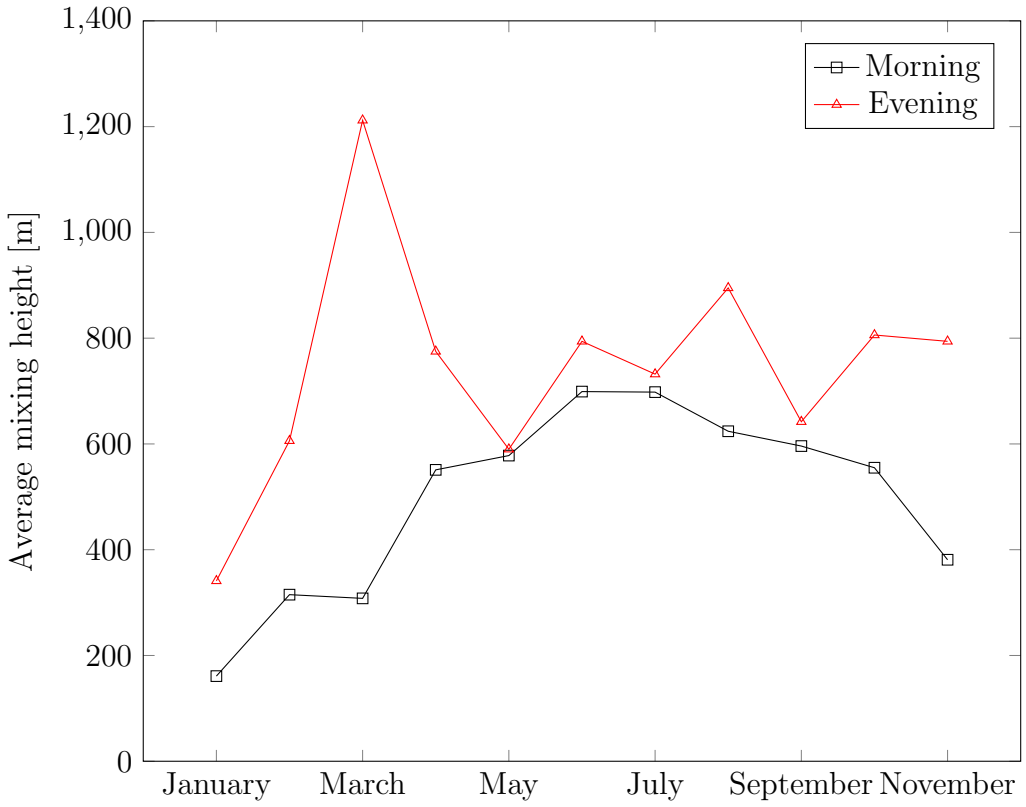
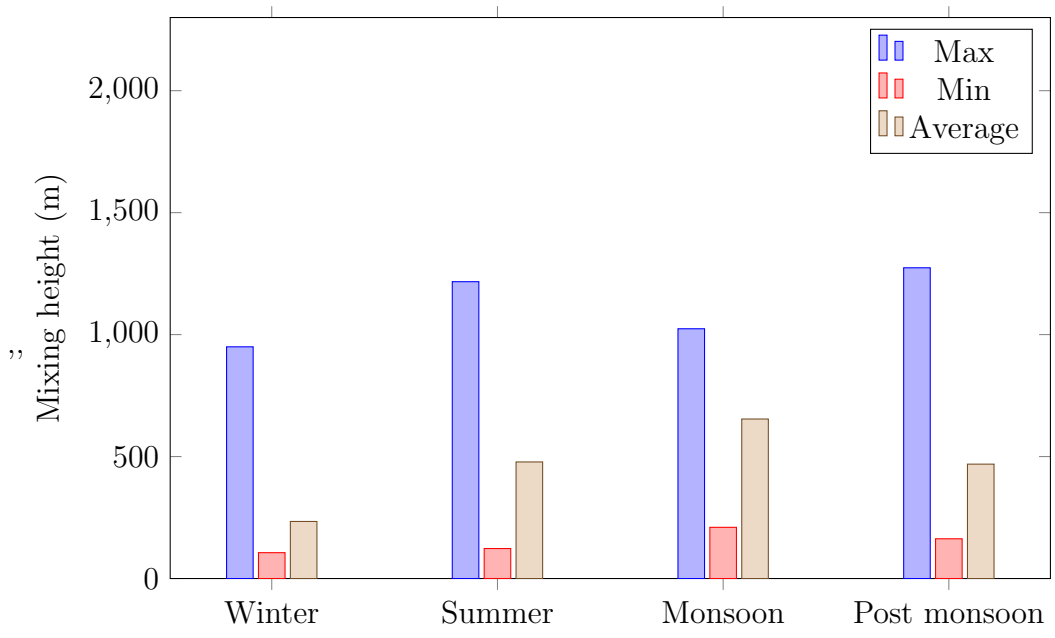


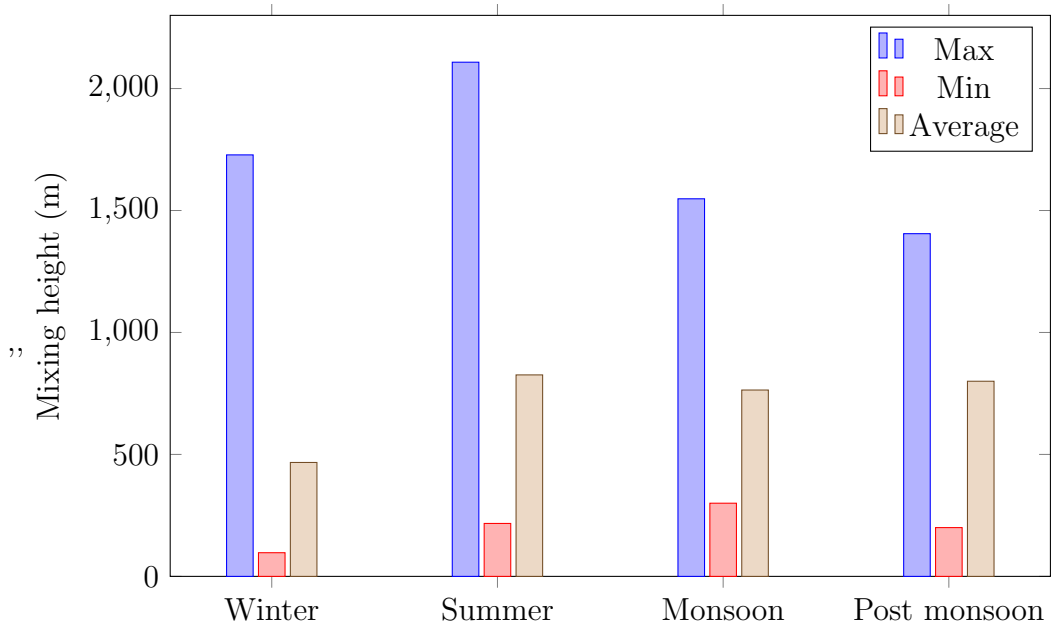
Figure 5.6: Monthly variation of mixing height

Seasonal variation

Seasonal maximum, minimum and average variation of mixing height for both morning and evening hours are shown in the figure fig 5.7a and fig 5.7b Seasonal variation of mixing height clearly shows that summer has outperformed all other seasons in terms of max morning and evening hours mixing height of 1216.8 meter and 2108.2 meter respectively. For average seasonal variation- max average value of 654.2 meter during monsoon for morning hours and 825.8 meter during summer for evening hours respectively was observed. Seasonal minimum value of mixing heights during morning hour was observed during winter (105.82m) and that for evening hours was also during winter (96.33m) which is the lowest for the entire year also.



(a) morning hours



(b) Evening hours

Figure 5.7: Seasonal variation of mixing height

5.1.4 Ventilation coefficient (VC)

Ventilation coefficient (VC) is calculated as the product of mixing height (MH) and mean wind speed (U).Hence,

$$VC = MH \times U \tag{5.1}$$

The seasonal variation of ventilation co efficient with respect to 24hour average wind speed for both of morning and evening time is calculated and shown in the fig 5.8, fig 5.9, fig 5.10 and fig 5.11. Seasonal duration is taken as per the seasons in the Assam such as Winter (January and February), Summer (March, April and May), Monsoon (June, July, August and September) and Post-monsoon (October, November and December).

Seasonal and monthly variation (fig 5.12) of ventilation coefficient for both morning and evening hours is shown in the following pages.

The table 5.1 shows the duration in consideration like morning and evening hour and along with seasons.

Table 5.1: Duration in consideration for different calculation

Time period	From	To
24hour average	00:00:00hr	23:59:59hr
12hour average (morning)	18:00:00hr	5:59:59hr
12hour average (evening)	6:00:00hr	17:59:59hr
Winter season	01-01-2019	28-02-2019
Summer season	01-03-2019	31-05-2019
Monsoon season	01-06-2019	30-09-2019
Post monsoon season	01-10-2019	30-11-2019

Seasonal variation of ventilation coefficient

Average seasonal variation of morning and evening ventilation coefficient as shown in the table 5.7 shows that the area has low assimilative capacity as per the NEERI classification.

Table 5.2: Seasonal variation of VC and corresponding assimilative capacity as per NEERI

Season	Morning average VC	Evening average VC	Assimilative capacity as per NEERI	Pollutant potential as per NEERI
Winter	490	950	Low	High
Summer	1230	2105	Low	High
Monsoon	1634	1890	Low	High
Post monsoon	775	1302	Low	High

Winter

Seasonal variation of ventilation coefficient for the winter season as shown in the fig 5.8 shows that during the morning hour

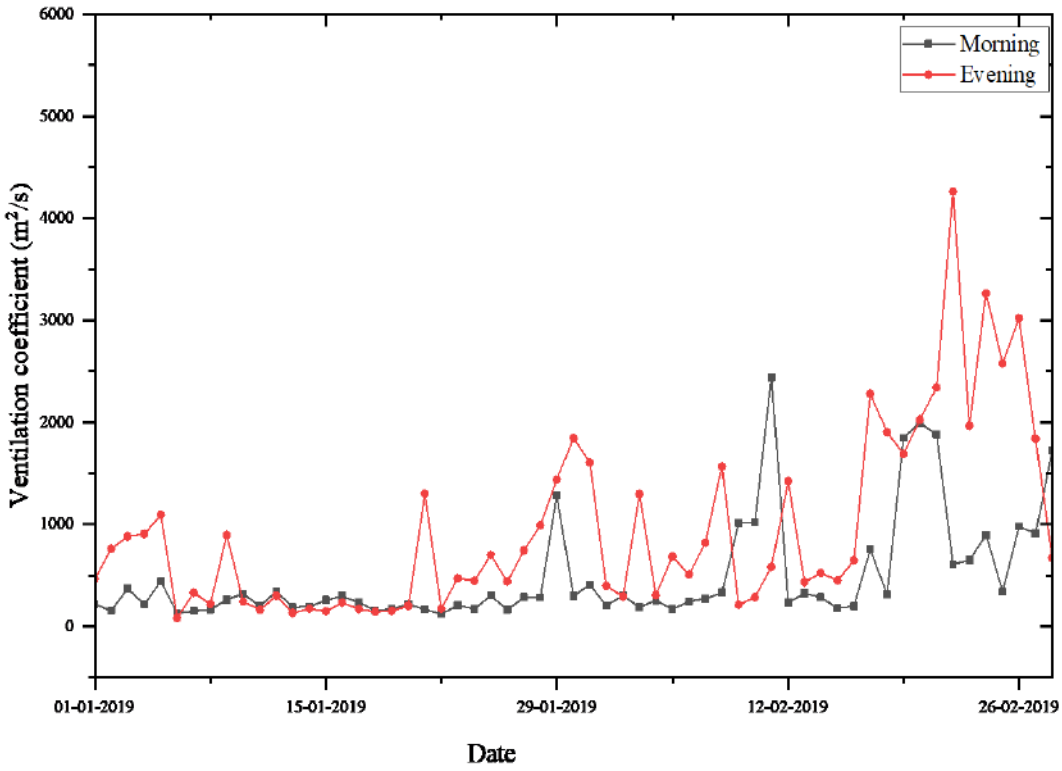


Figure 5.8: Variation of VC for winter

a max value of 2442 m^2/s is observed while the min value reads 123.5 m^2/s that for evening hours a max value of 4262 m^2/s is observed while minimum value reads 84 m^2/s. Hence, for the entire season the ventilation co efficient value varies from as low as 84 m^2/s to a significant high 4262 m^2/s.

Summer

Seasonal variation of ventilation coefficient for the summer season as shown in the fig 5.9 shows that during the morning hour

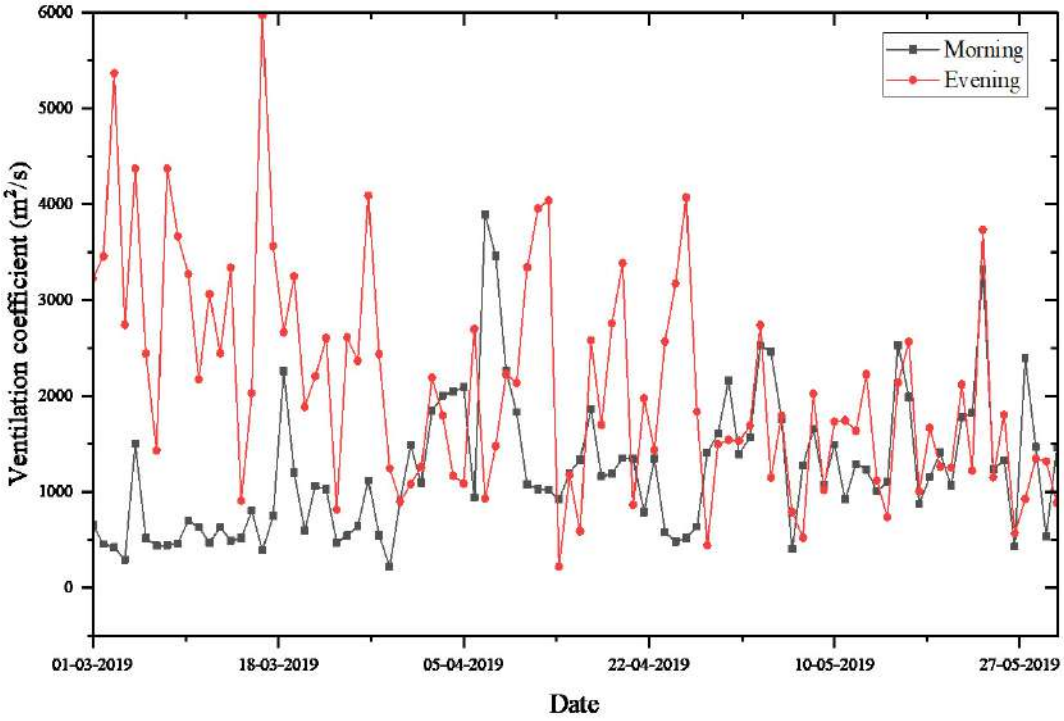


Figure 5.9: Variation of VC for summer

a max value of 3893 m^2/s is observed while the min value reads 214 m^2/s that for evening hours a max value of 5977 m^2/s is observed while minimum value reads 219 m^2/s. Hence, for the entire season the ventilation co efficient value varies from as low as 214 m^2/s to a significant high 5977 m^2/s.

Monsoon

Seasonal variation of ventilation coefficient for the monsoon season as shown in the fig 5.10 shows that during the morning hour

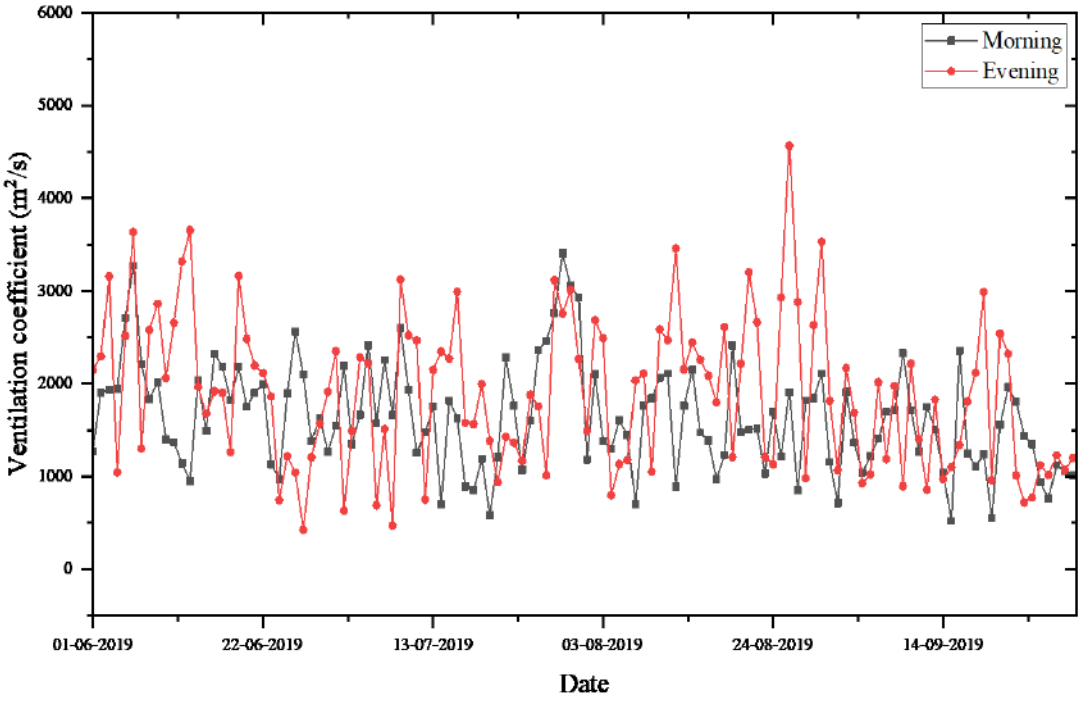


Figure 5.10: Variation of VC for monsoon

a max value of 3464 m^2/s is observed while the min value reads 516 m^2/s that for evening hours a max value of 4565 m^2/s is observed while minimum value reads 424 m^2/s. Hence, for the entire season the ventilation co efficient value varies from as low as 424 m^2/s to a significant high 4565 m^2/s.

Post monsoon

Seasonal variation of ventilation coefficient for the post monsoon season as shown in the fig 5.11 shows that during the morning hour a max value of 2212 m^2/s is

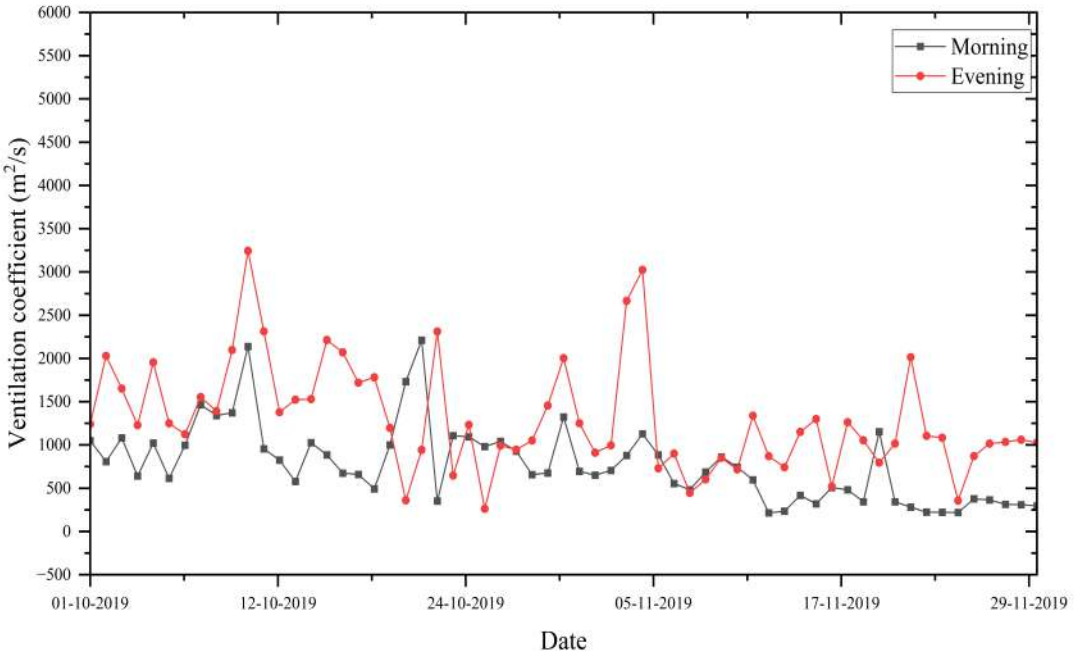


Figure 5.11: Variation of VC for post monsoon

observed while the min value reads 215 m^2/s that for evening hours a max value of 3244 m^2/s is observed while minimum value reads 263 m^2/s. Hence, for the entire season the ventilation co efficient value varies from as low as 215 m^2/s to a significant high 3244 m^2/s.

Monthly variation of ventilation coefficient

Monthly average variation of ventilation coefficients for morning and evening hours as shown in the fig 5.12 shows that

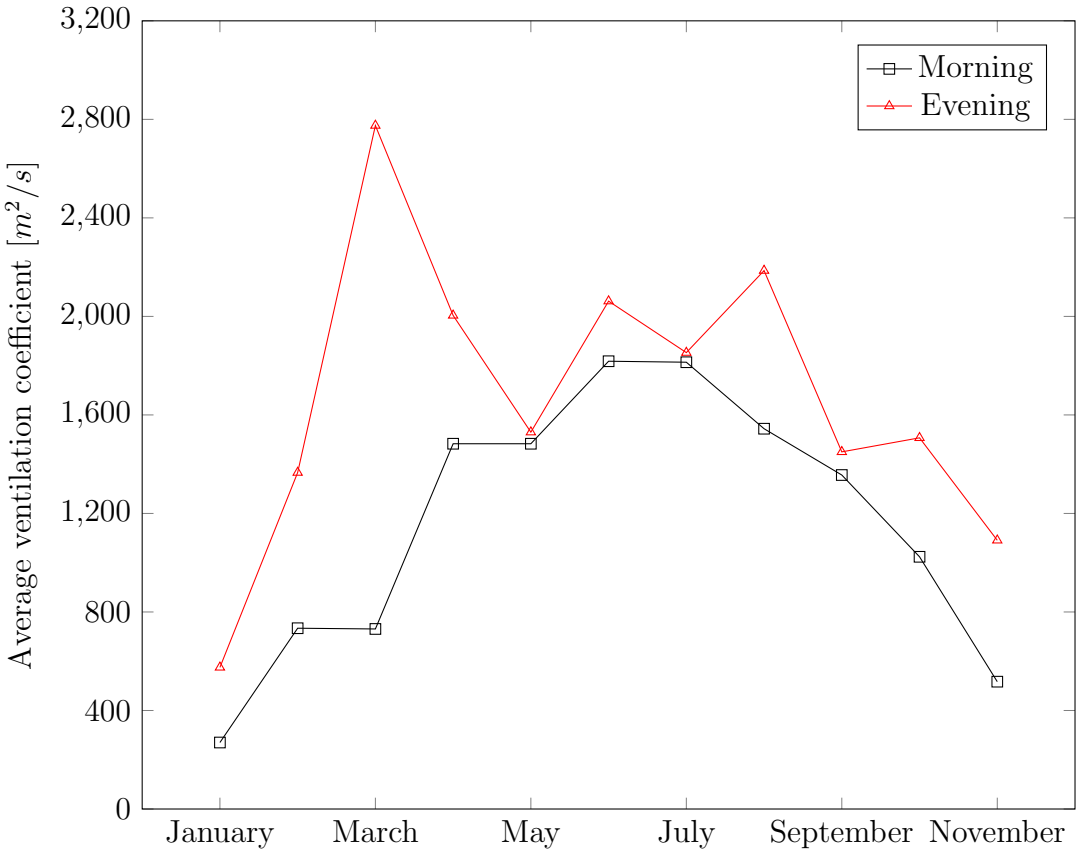


Figure 5.12: Monthly variation of mixing height

for morning hours max average value of 1818 m^2/s is observed during June while a min average value of 270 m^2/s is observed during January. Similarly, for the evening hour it has been observed that a max value of 2775 m^2/s is prevailed during march while a min value of 578.2 m^2/s is prevailed during January.

5.2 Air quality modelling

Dispersion modelling for two major air pollutants namely sulphur dioxide and oxides of nitrogen from the refinery is performed for all four seasons and along with that for annual variation. The observed results and model input data are shown in the following sections. The variation of predicted GLC at the discrete receptors level is also shown in a comparative analysis at the end of this section. The input data for the point sources consists of location of the stacks, release height of the stacks, stack internal diameter at release height, exit gas velocity, exit gas temperature and emission rate of pollutants from the stacks. Among all the parameters only emission rate of pollutants and exit gas velocity varies seasonally and also based on the pollutants. Hence, here fixed parameters are shown in the table 5.3 and the variable parameters are shown in their respective section.

Table 5.3: Fixed input parameter for the point sources for both the pollutants. (Source:EIA report of refinery)

Point sources	UTM East (m)	UTM North (m)	Release height (m)	Stack internal diameter (m)	Exit gas tempera-ture (K)
Stack1	758876.5	3031890.07	60	1.5	555
Stack2	758875.94	3031916.45	60	1.5	525
Stack3	758878.16	3031942.67	80	3.0	575
Stack4	758853.64	3031842.69	50	1.5	560
Stack5	758964.76	3032154.5	50	2.5	573
Stack6	759114.97	3031925.92	50	2.5	585
Stack7	758813.59	3031916.25	50	1.5	550
Stack8	759111.29	3031983.28	50	2.5	590
DG set1	758782.83	3031915.16	50	.12	560
DG set2	758791.7	3031912.46	50	.12	540

The fig 5.13a shows the located position of the point sources in the study domain where fig 5.13b shows the actual position of the point sources.The running time of the stacks are not available, hence it is assumed that the stacks were running throughout year and a constant emission factor of 1 was chosen for all the point sources as per the similar assumptions made by the D.Gibson and co authors for the dispersion modelling of pollutants (Gibson *et al.* 2013).



(a) Located position of the stacks



(b) Actual position of the stacks. Source:https://www.iocl.com/aboutus/DigboiRefinery.aspx

Figure 5.13: Position of the stacks

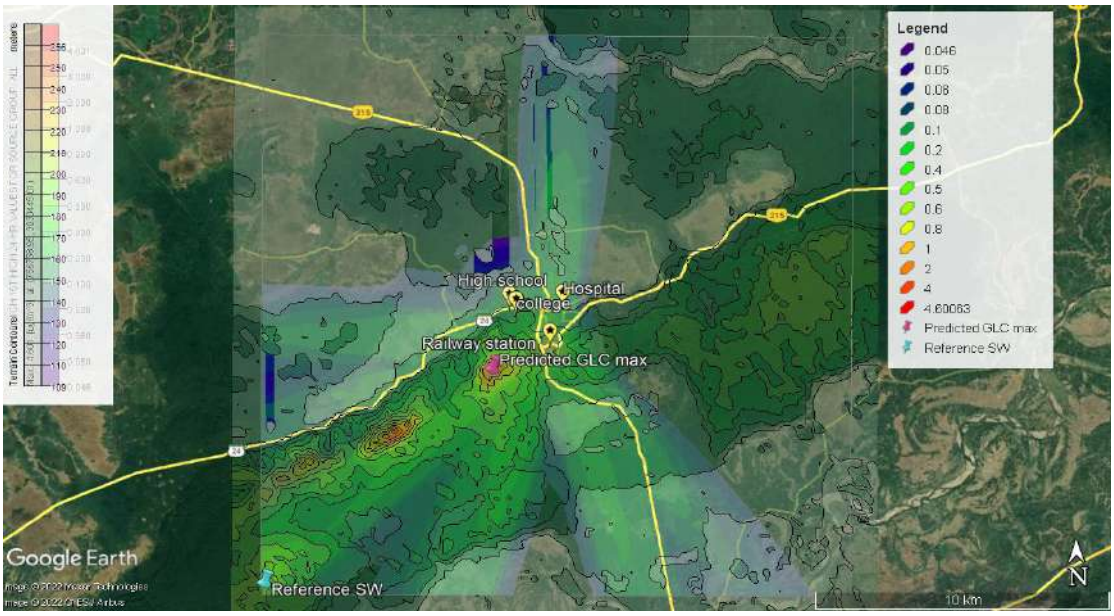
5.2.1 Sulphur dioxide (SO_2)

The model simulation is done for all the four seasons and as well as for annual variation of the pollutant. The input data of emission rate and exit gas velocity for the different point sources and for different seasons are shown in the table 5.4.

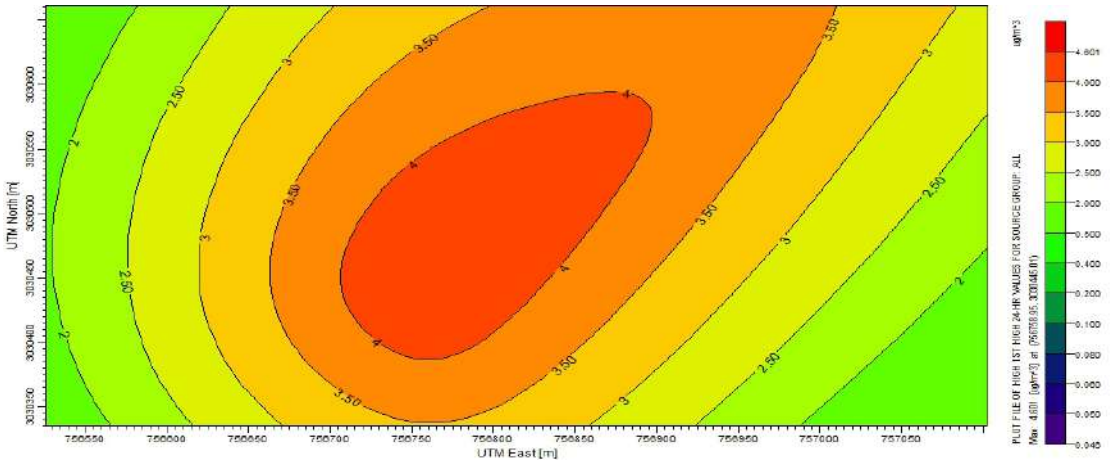
Table 5.4: Variable stack input data for SO_2. (Source:EIA report of refinery)

Emission rate (g/s)					
Stacks	Winter	Summer	Monsoon	Post monsoon	Annual
Stack1	0.139	0.150	0.150	0.139	0.139
Stack2	0.175	0.161	0.150	0.136	0.175
Stack3	0.389	0.753	0.297	0.831	0.831
Stack4	0.100	0.167	0.161	0.103	0.167
Stack5	0.172	0.303	0.322	0.192	0.322
Stack6	0.144	0.403	0.408	0.103	0.410
Stack7	0.103	0.100	0.100	0.100	0.103
Stack8	0.100	0.203	0.158	0.064	0.203
DG1	0.0622				
DG2	0.0393				
Exit gas velocity (m/s)					
Stacks	Winter	Summer	Monsoon	Post monsoon	Annual
Stack1	2.124	2.345	2.513	2.122	2.513
Stack2	2.124	2.345	2.513	2.122	2.513
Stack3	0.531	0.586	0.628	0.530	0.628
Stack4	2.124	2.345	2.513	2.122	2.513
Stack5	0.765	0.844	0.904	0.764	0.904
Stack6	0.765	0.844	0.904	0.764	0.904
Stack7	2.124	2.345	2.513	2.122	2.513
Stack8	0.765	0.844	0.904	0.764	0.904
DG1	22.3				
DG2	16.1				

Winter



(a) Variation over the study domain

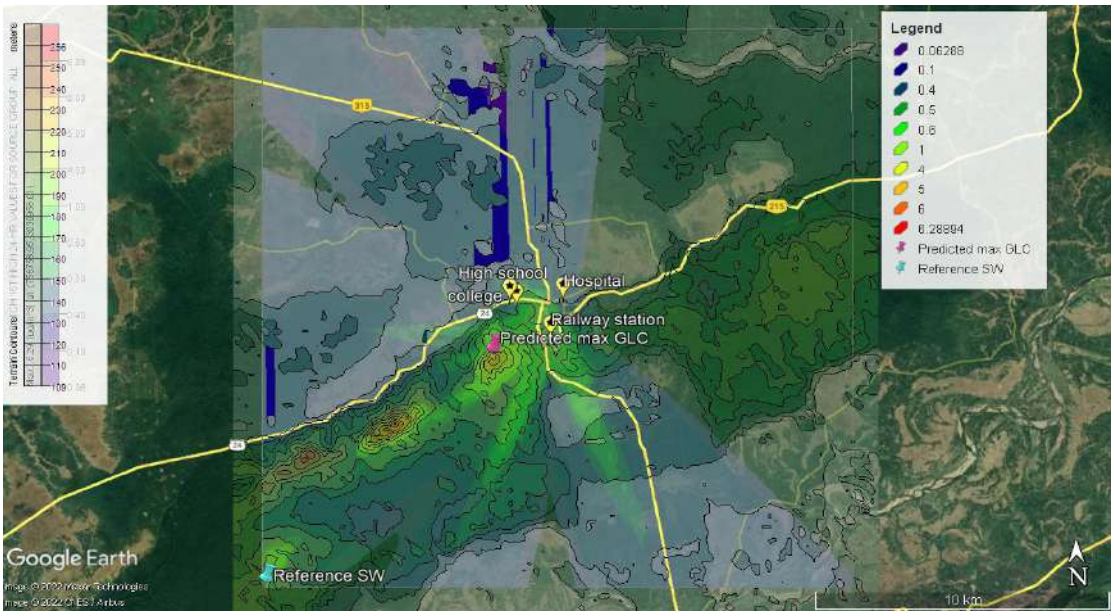


(b) Zoomed view of critical concentration location

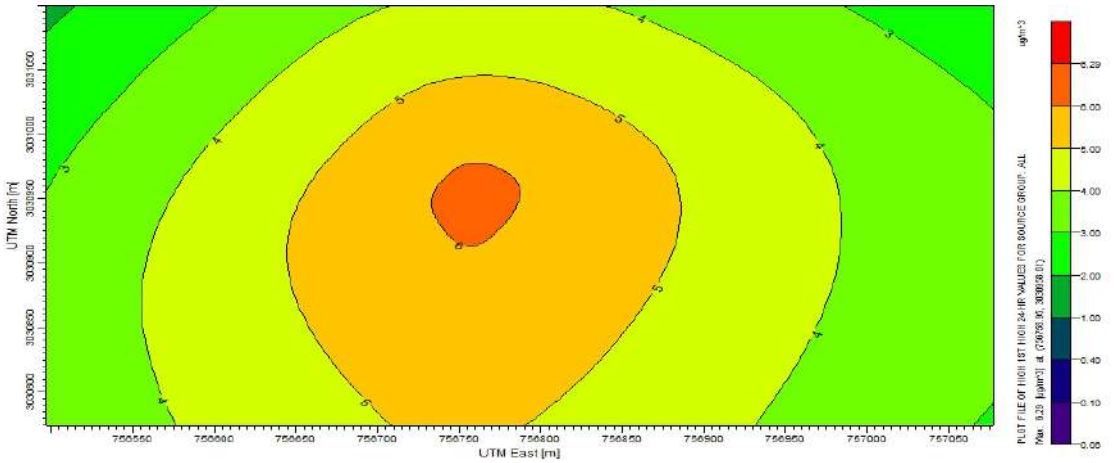
Figure 5.14: Winter SO_2 variation of predicted concentration over the study domain

Predicted GLC's over the study area is shown in the fig 5.14a max GLC value of 4.6 $\mu g/m^3$ is observed at a distance of 2.7 km in the down wind direction as shown in the fig 5.14b during the winter season and among all discrete receptors max predicted GLC value of 1 $\mu g/m^3$ is observed near the railway station area.

Summer



(a) Variation over the study domain

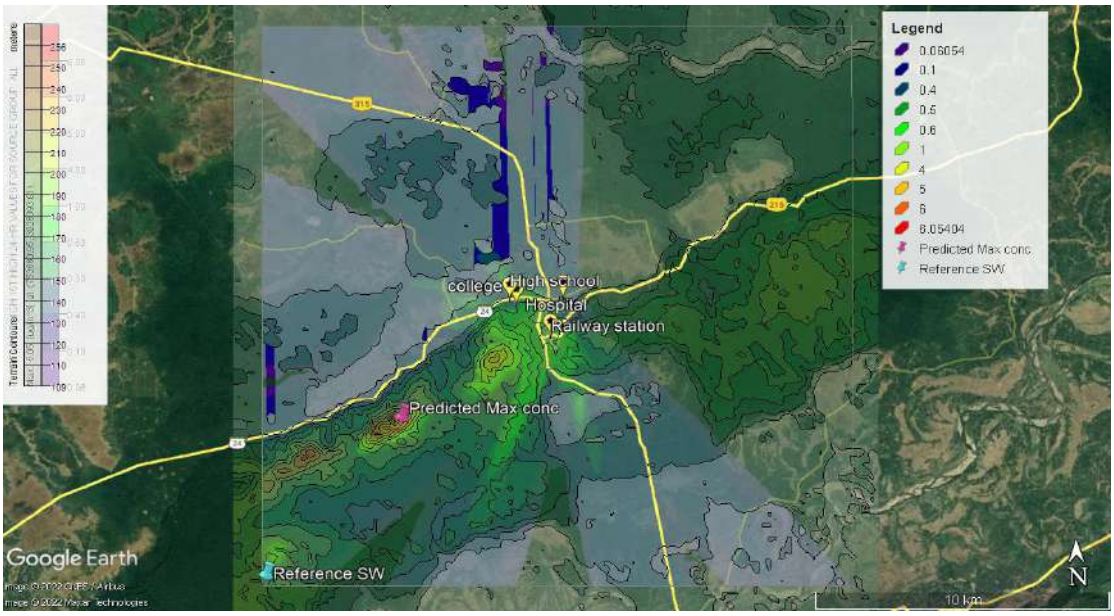


(b) Zoomed view of critical concentration location

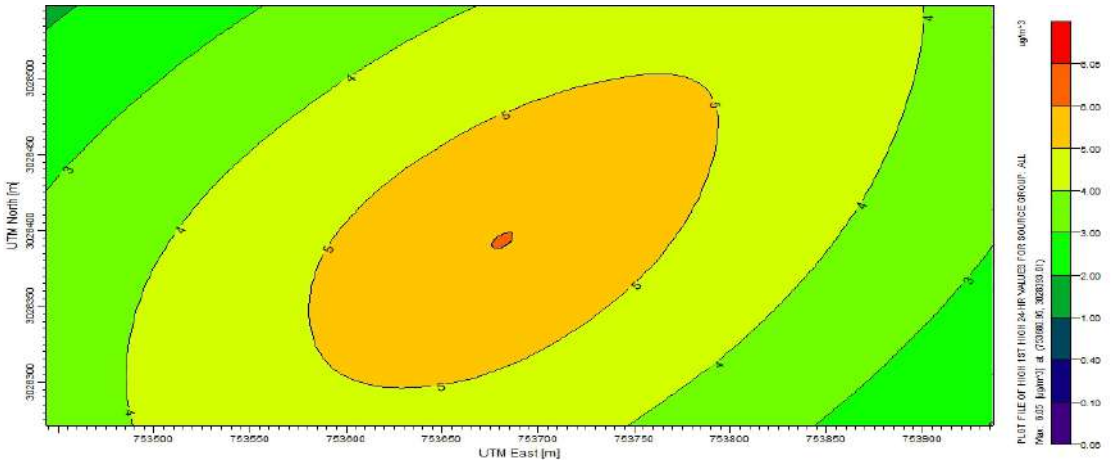
Figure 5.15: Summer SO_2 variation of predicted concentration over the study domain

Predicted GLC's over the study area is shown in the fig 5.15a, max GLC value of 6.3 $\mu g/m^3$ is observed at a distance of 2.65 km in the down wind direction as shown in the fig 5.15b during the summer season and among all discrete receptors max predicted GLC value of 1.9 $\mu g/m^3$ is observed near the railway station area.

Monsoon



(a) Variation over the study domain

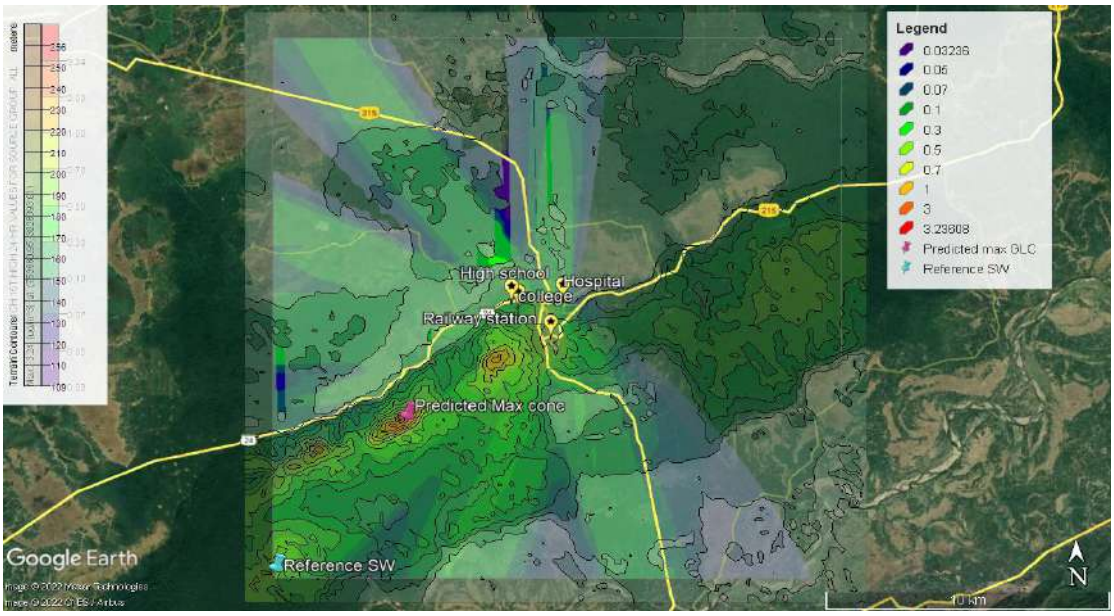


(b) Zoomed view of critical concentration location

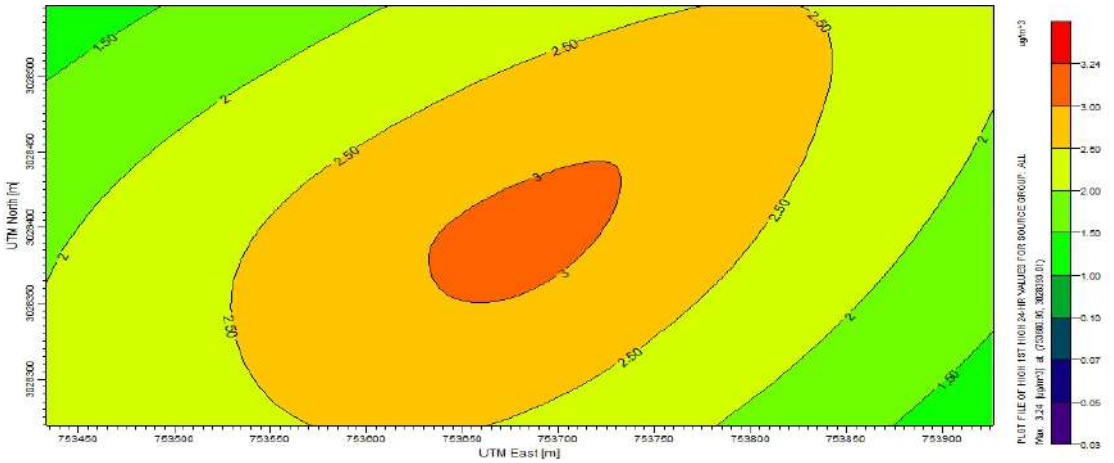
Figure 5.16: Monsoon SO_2 variation of predicted concentration over the study domain

Predicted GLC's over the study area is shown in the fig 5.16a, max GLC value of 6.05 $\mu g/m^3$ is observed at a distance of 6.5 km in the down wind direction as shown in the fig 5.16b during the monsoon season and among all discrete receptors max predicted GLC value of 1.83 $\mu g/m^3$ is observed near the railway station area.

Post monsoon



(a) Variation over the study domain

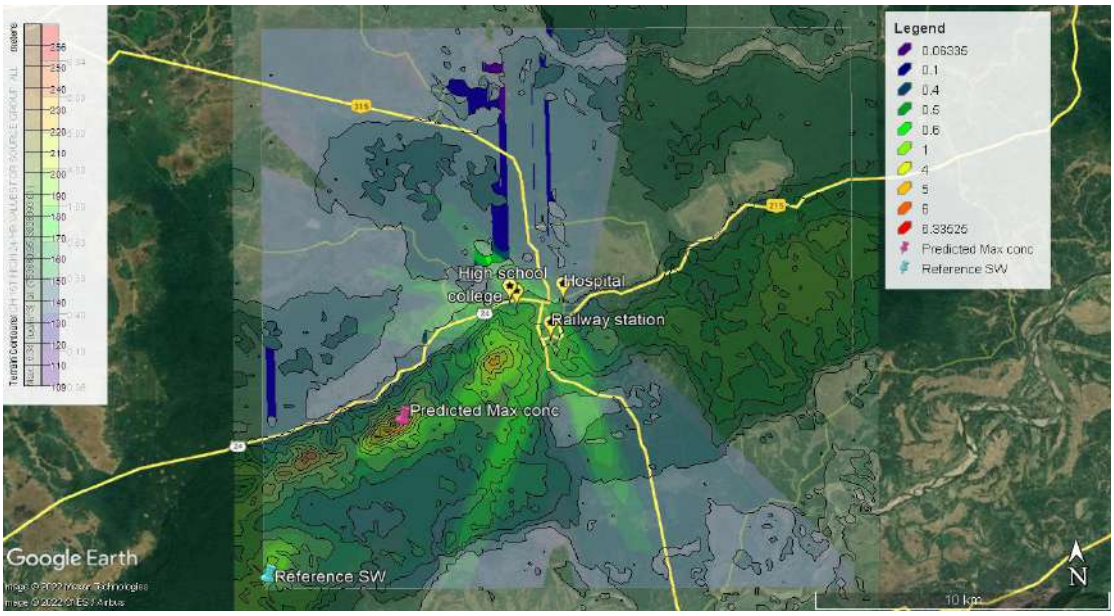


(b) Zoomed view of critical concentration location

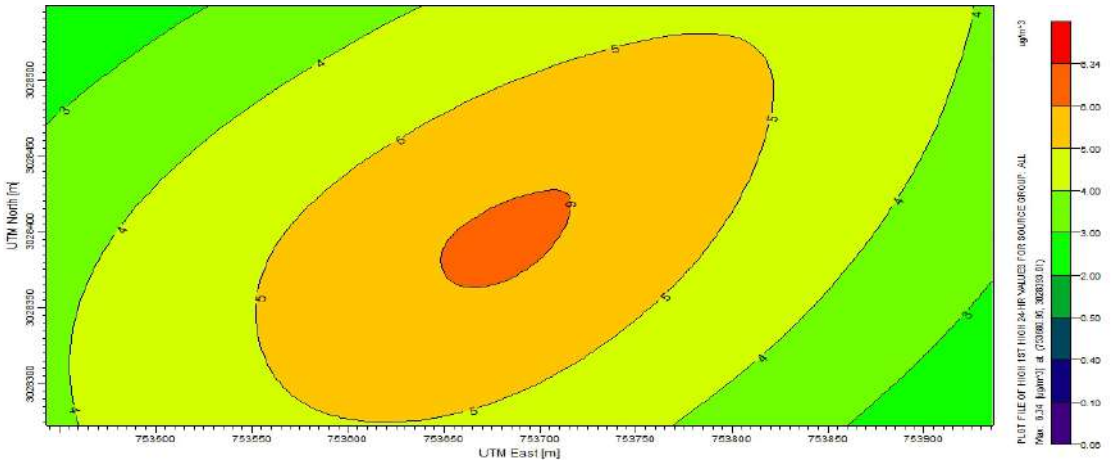
Figure 5.17: Post monsoon SO_2 variation of predicted concentration over the study domain

Predicted GLC's over the study area is shown in the fig 5.17a, max GLC value of 3.23 $\mu g/m^3$ is observed at a distance of 6.5 km in the down wind direction as shown in the fig 5.17b during the post monsoon season and among all discrete receptors max predicted GLC value of 1.32 $\mu g/m^3$ is observed near the railway station area.

Annual



(a) Variation over the study domain



(b) Zoomed view of critical concentration location

Figure 5.18: Annual SO_2 variation of predicted concentration over the study domain

Predicted GLC's over the study area is shown in the fig 5.18a, max GLC value of 6.34 $\mu g/m^3$ is observed at a distance of 6.5 km in the down wind direction as shown in the fig 5.18b during the winter season and among all discrete receptors max predicted GLC value of 2.2 $\mu g/m^3$ is observed near the railway station area.

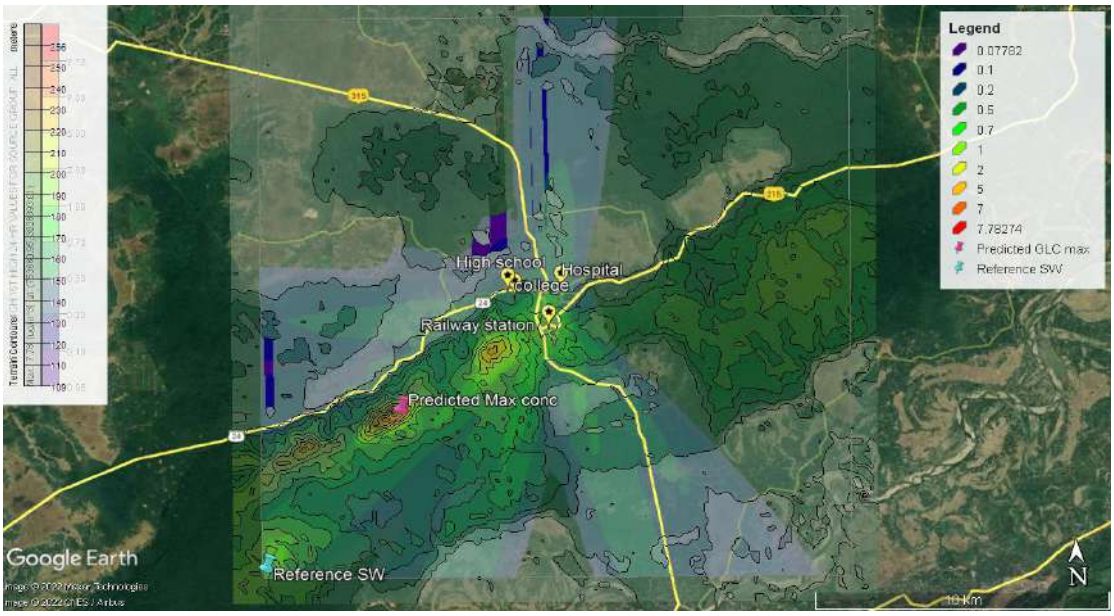
5.2.2 Oxides of nitrogen (NO_x)

The model simulation is done for all the four seasons and as well as for annual variation of the pollutant. The input data of emission rate and exit gas velocity for the different point sources and for different seasons are shown in the table 5.5.

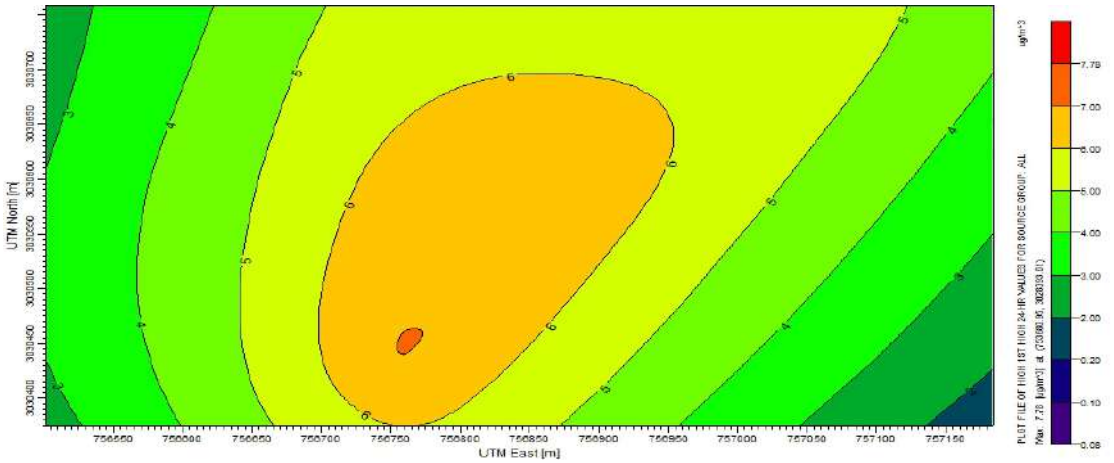
Table 5.5: Variable stack input data for NO_x. (Source:EIA report of refinery)

Emission rate (g/s)					
Stacks	Winter	Summer	Monsoon	Post monsoon	Annual
Stack1	0.278	0.428	0.505	0.350	0.505
Stack2	0.330	0.530	0.382	0.328	0.530
Stack3	0.350	0.302	0.534	0.941	0.941
Stack4	0.197	0.223	0.364	0.311	0.364
Stack5	.300	0.235	0.425	0.331	0.425
Stack6	0.342	0.167	0.383	0.350	0.383
Stack7	0.281	0.100	0.297	0.253	0.297
Stack8	0.350	0.161	0.430	0.375	0.430
DG1	0.0697				
DG2	0.0438				
Exit gas velocity (m/s)					
Stacks	Winter	Summer	Monsoon	Post monsoon	Annual
Stack1	2.124	2.345	2.513	2.122	2.513
Stack2	2.124	2.345	2.513	2.122	2.513
Stack3	0.531	0.586	0.628	0.530	0.628
Stack4	2.124	2.345	2.513	2.122	2.513
Stack5	0.765	0.844	0.904	0.764	0.904
Stack6	0.765	0.844	0.904	0.764	0.904
Stack7	2.124	2.345	2.513	2.122	2.513
Stack8	0.765	0.844	0.904	0.764	0.904
DG1	22.3				
DG2	16.1				

Winter



(a) Variation over the study domain

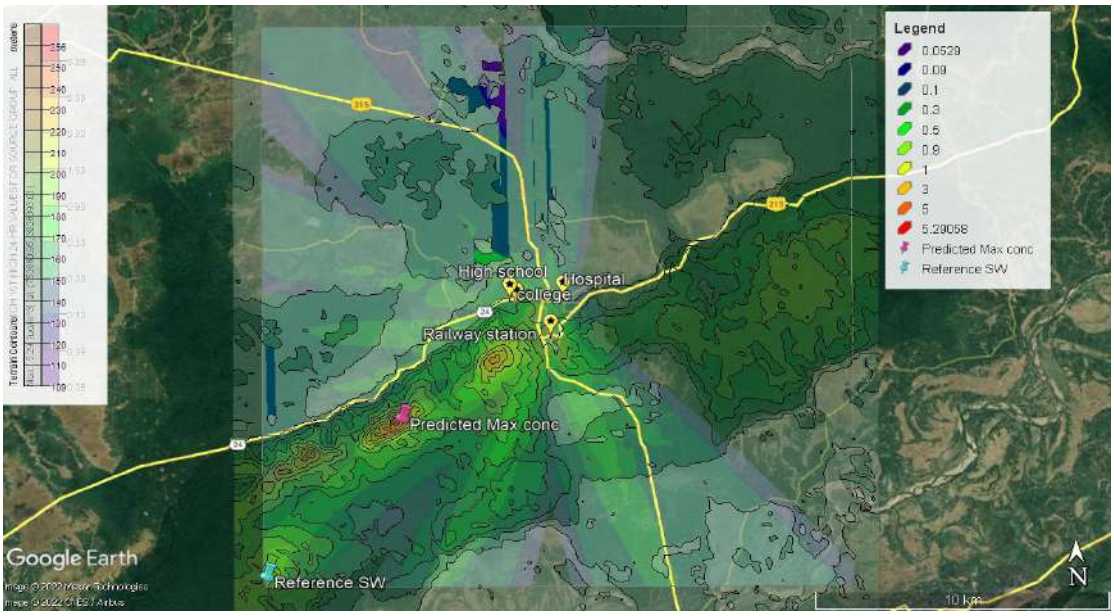


(b) Zoomed view of critical concentration location

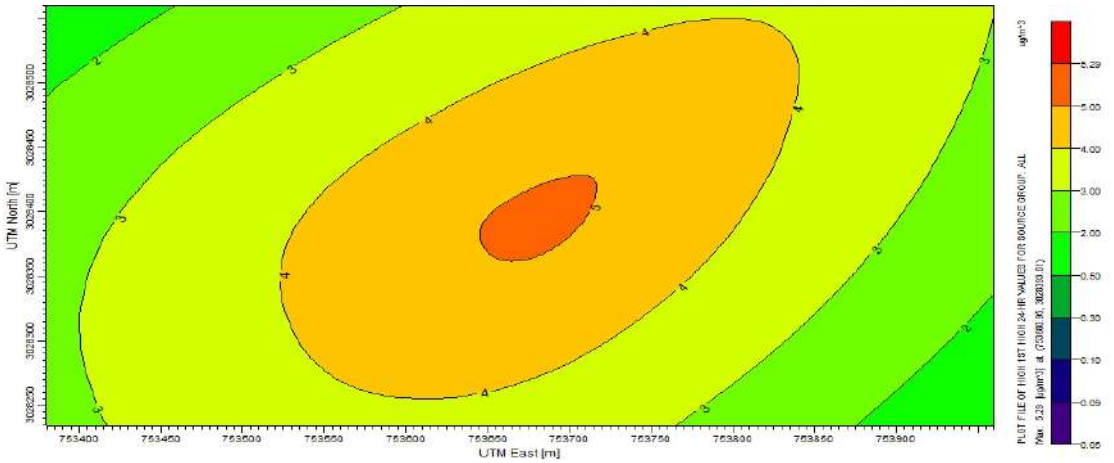
Figure 5.19: Winter NO_x variation of predicted concentration over the study domain

The predicted GLC for the study area is shown in the fig 5.19a and max GLC of 7.78 $\mu g/m^3$ was predicted at a distance of 6.5 km from the point sources in the down wind direction as shown in the fig 5.19b and that for discrete receptors at railway station area predicted GLC of 1.89 $\mu g/m^3$ prevails throughout the season.

Summer



(a) Variation over the study domain

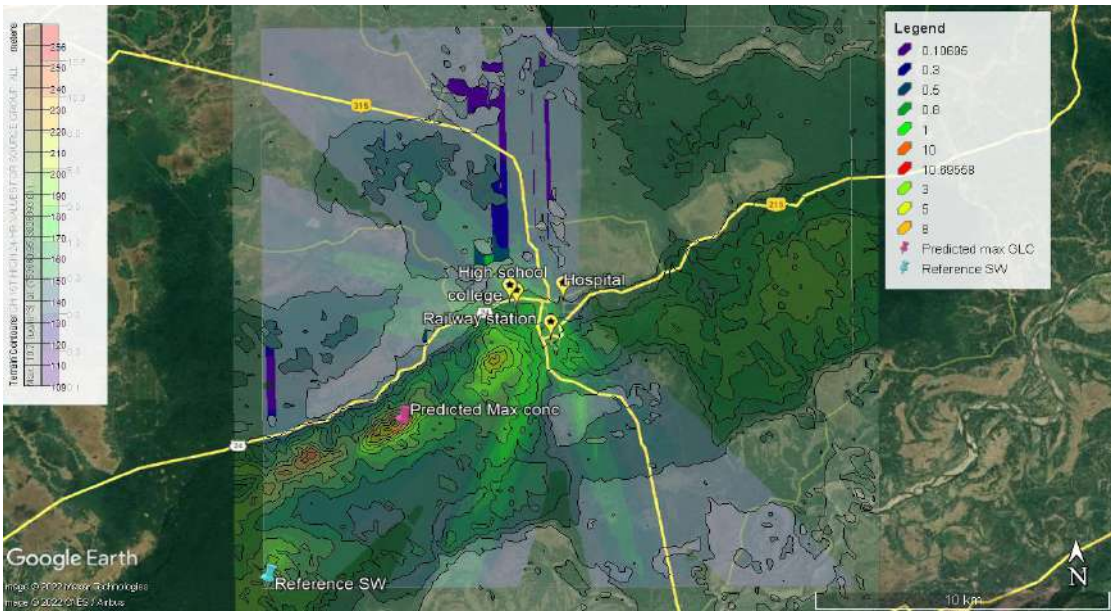


(b) Zoomed view of critical concentration location

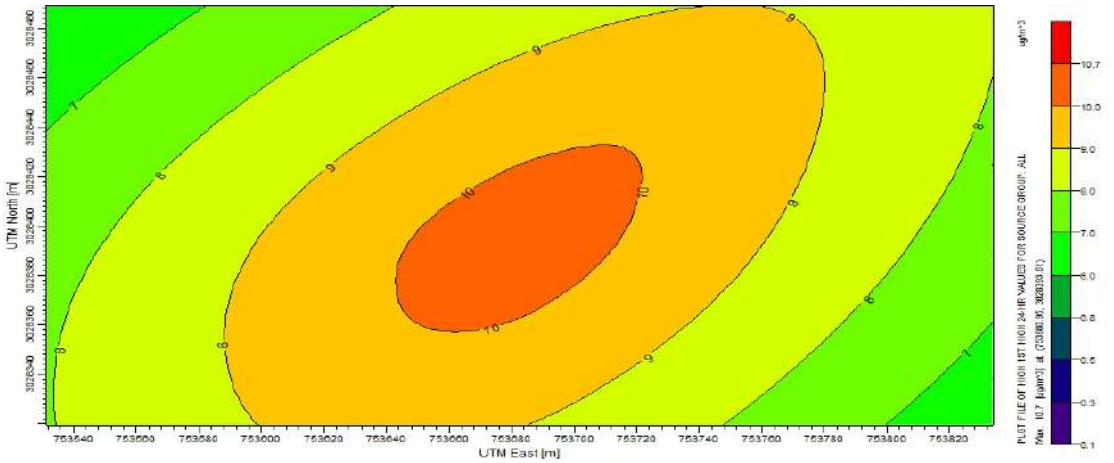
Figure 5.20: Summer NO_x variation of predicted concentration over the study domain

For the summer season predicted GLC's over the study area is shown in the fig 5.20a max predicted GLC was 5.29 $\mu g/m^3$ in the south west direction from the point sources at a distance of 6.5 km in the down wind direction as shown in the fig 5.20b and that for discrete receptors it was at the railway station area and the corresponding concentration is 1.8 $\mu g/m^3$.

Monsoon



(a) Variation over the study domain

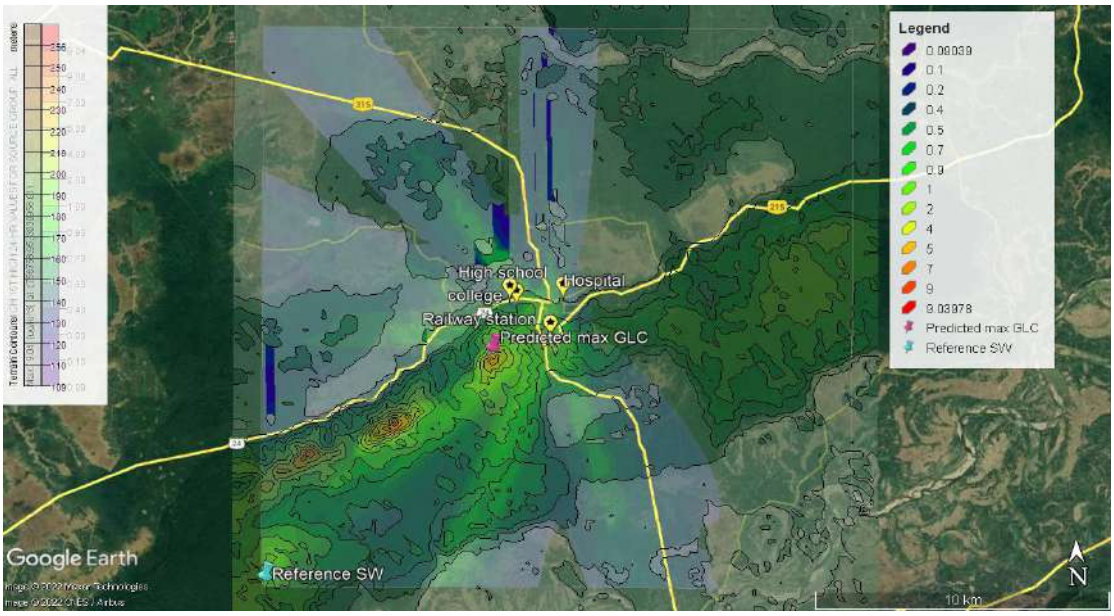


(b) Zoomed view of critical concentration location

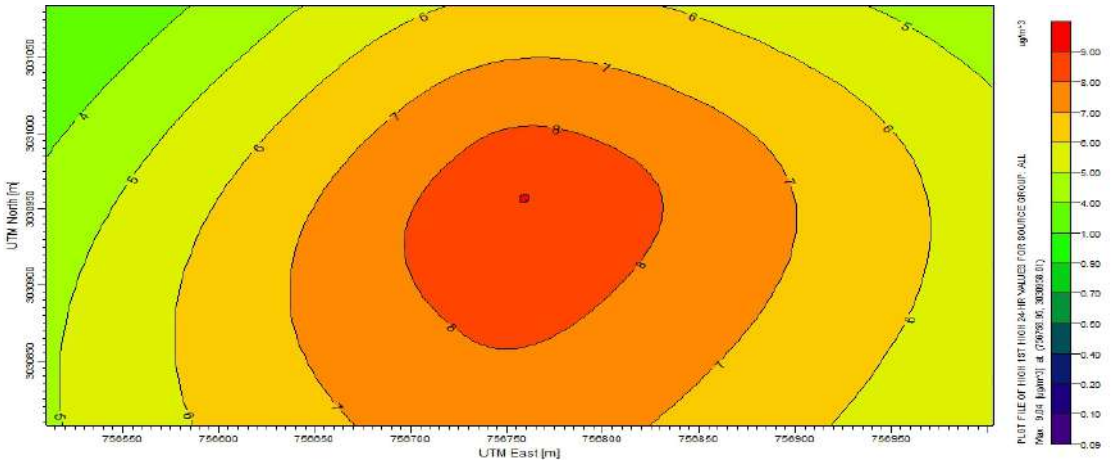
Figure 5.21: Monsoon NO_x variation of predicted concentration over the study domain

Model prediction for the monsoon season as shown in the fig 5.21a shows that max GLC of 10.7 $\mu g/m^3$ was observed at a distance of 6.5 km from the point sources in the down wind direction as shown in the fig 5.21b. Among all discrete receptors in the railway station area max GLC of 3.09 $\mu g/m^3$ was predicted during the entire season.

Post monsoon



(a) Variation over the study domain

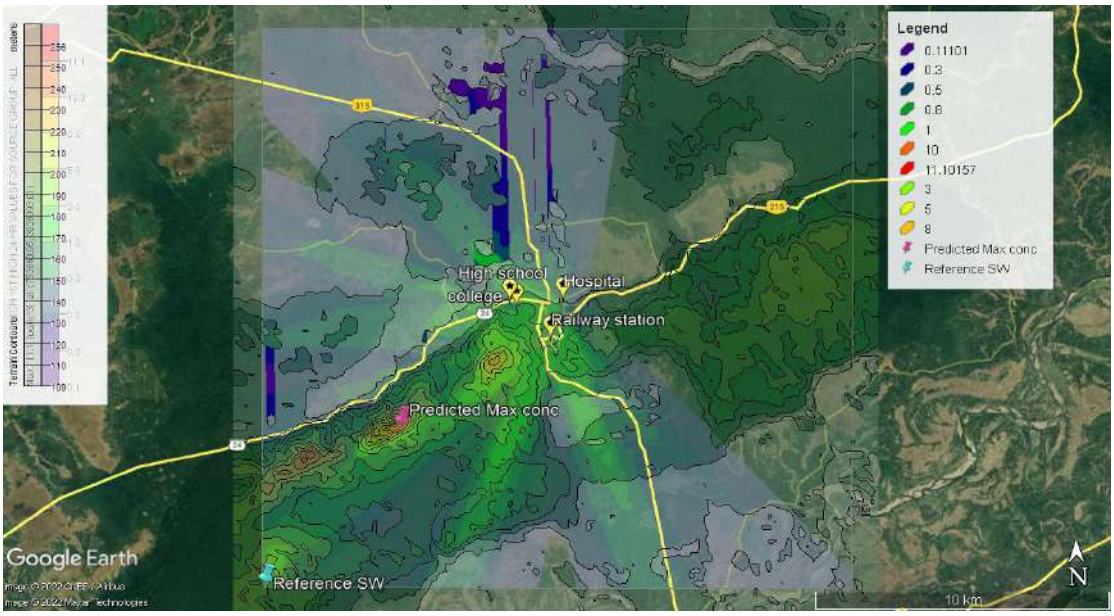


(b) Zoomed view of critical concentration location

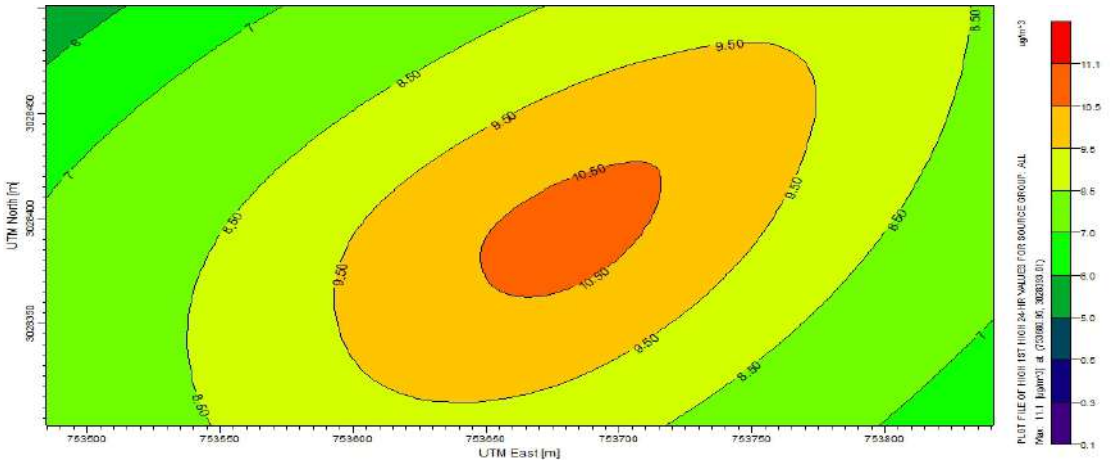
Figure 5.22: Post monsoon NO_x variation of predicted concentration over the study domain

Predicted GLC's over the study area is shown in the fig 5.22a and max GLC of 9.04 $\mu g/m^3$ was observed at a distance 2.65 km in the down wind direction from the point sources during the post monsoon season as shown in the fig 5.22b and that for discrete receptors max GLC is predicted value of 2.61 $\mu g/m^3$ for the railway station area.

Annual



(a) Variation over the study domain



(b) Zoomed view of critical concentration location

Figure 5.23: Annual NO_x variation of predicted concentration over the study domain

Predicted annual GLC's over the study area is shown in the fig 5.23a max GLC value of 11.1 $\mu g/m^3$ is observed at a distance 6.5 km in the down wind direction for the entire year as shown in the fig 5.23b and among all discrete receptors in the railway station max GLC value of 3.47 $\mu g/m^3$ is predicted for the entire year.

5.2.3 Comparison of model results

Initially the model was run for the existing emission rate from the point sources and the predicted GLC's at discrete receptors and max GLC over the study domain is noted for both the pollutants and then the emission rate is increased uniformly for both the pollutants until the max predicted GLC over the study domain exceeds. This process is repeated for all the seasons and the results are tabulated for both the pollutants.

Table 5.6: Comparison of predicted GLC's at discrete receptors for SO_2

Present emission rate (g/s)					
		Predicted concentrations ($\mu g/m^3$)			
Season	Emission rate	Railway station	College area	High school	Hospital area
Winter	1.44	1	0.19	0.14	0.25
Summer	2.31	1.9	0.5	0.4	0.3
Monsoon	1.86	1.83	0.44	0.3	0.24
Post monsoon	1.78	1.32	0.22	0.15	0.17
Annual	2.46	2.2	0.56	0.4	0.45
Max emission rate (g/s) corresponding to 80 $\mu g/m^3$ concentration					
		Predicted concentrations ($\mu g/m^3$)			
Season	Emission rate	Railway station	College area	High school	Hospital area
Winter	25	17.64	3.33	2.44	4.4
Summer	29.34	23.8	5.66	4.85	3.63
Monsoon	24.6	24.3	5.9	4	3.2
Post monsoon	44.2	33	5.6	3.7	4.24
Annual	31.1	28	7.11	5.1	5.74

For the max available assimilative potential the predicted concentrations at discrete receptors shows that for all the seasons and for both the pollutants as shown in table 5.6 and table 5.7 shows near the railway station area has witnessed the max pollutant concentrations.Though at existing pollution load of SO_2 railway station area has witnessed the max pollution concentration during the summer season but that for max available pollution potential, the area has witnessed the max pollutant concentration during the post monsoon season.

Similarly, for NO_x at existing pollution load max concentration was observed during monsoon season but that for max pollution load the max concentration is observed during summer for the railway station area.

But for uniform receptors, the receptor which has witnessed the max pollutant concentration during any season for any particular pollutant - the same will also be witnessing the max permissible value for the max pollutant concentration also.

At all the receptors predicted concentrations are well below the NAAQS standard for both SO_2 and NO_x i.e 50 $\mu g/m^3$ for industrial,residential and rural areas and 80 $\mu g/m^3$ for other area for SO_2 and that for NO_x is 40 $\mu g/m^3$ for industrial,residential and rural areas and 80 $\mu g/m^3$ for other area (CPCB 2009).

Table 5.7: Comparison of predicted GLC's at discrete receptors for NO_x

Present emission rate (g/s)					
		Predicted concentrations ($\mu g/m^3$)			
Season	Emission rate	Railway station	College area	High school	Hospital area
Winter	2.54	1.89	0.33	0.24	0.48
Summer	2.2	1.8	0.5	0.4	0.2
Monsoon	3.44	3.09	0.76	0.51	0.41
Post monsoon	3.35	2.61	0.42	0.28	0.42
Annual	4	3.47	0.86	0.67	0.71
Max emission rate (g/s)					
		Predicted concentrations ($\mu g/m^3$)			
Season	Emission rate	Railway station	College area	High school	Hospital area
Winter	26.2	19.48	3.42	2.51	5
Summer	33.5	26.5	7.27	6.12	3.58
Monsoon	25.8	23.3	5.7	3.9	3.1
Post monsoon	29.7	23.15	3.74	2.48	3.68
Annual	28.84	24.6	6.17	4.9	5.1

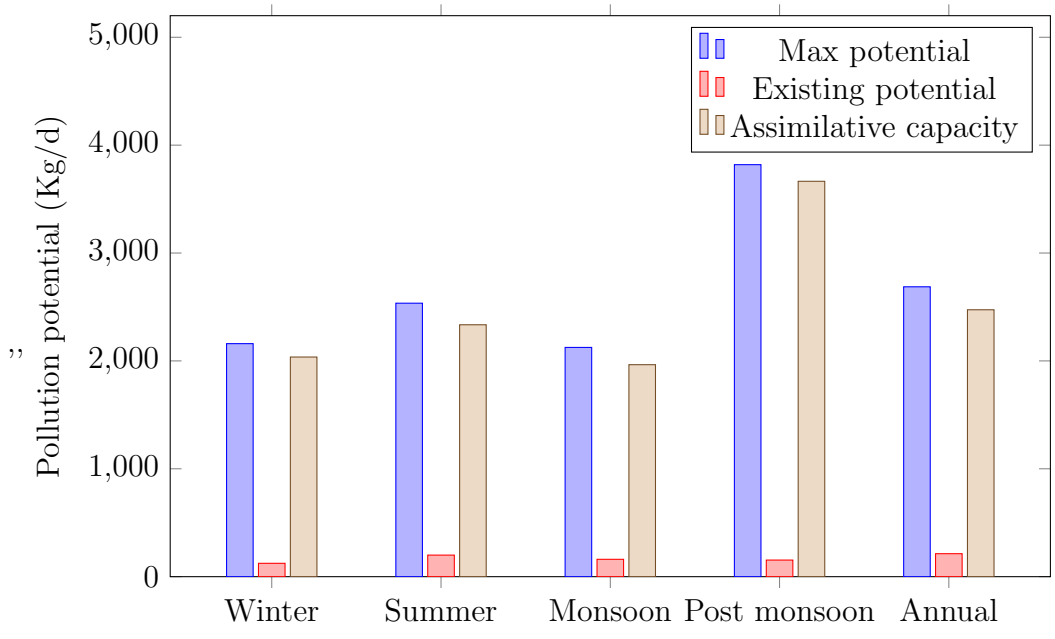
5.2.4 Assimilative capacity

Assimilative capacity is calculated as the difference between the max available pollution potential of that region (emission rate for which the max predicted concentration touched the permissible limits as per NAAQS for that pollutant) and the present pollution load of that region for that particular pollutant.

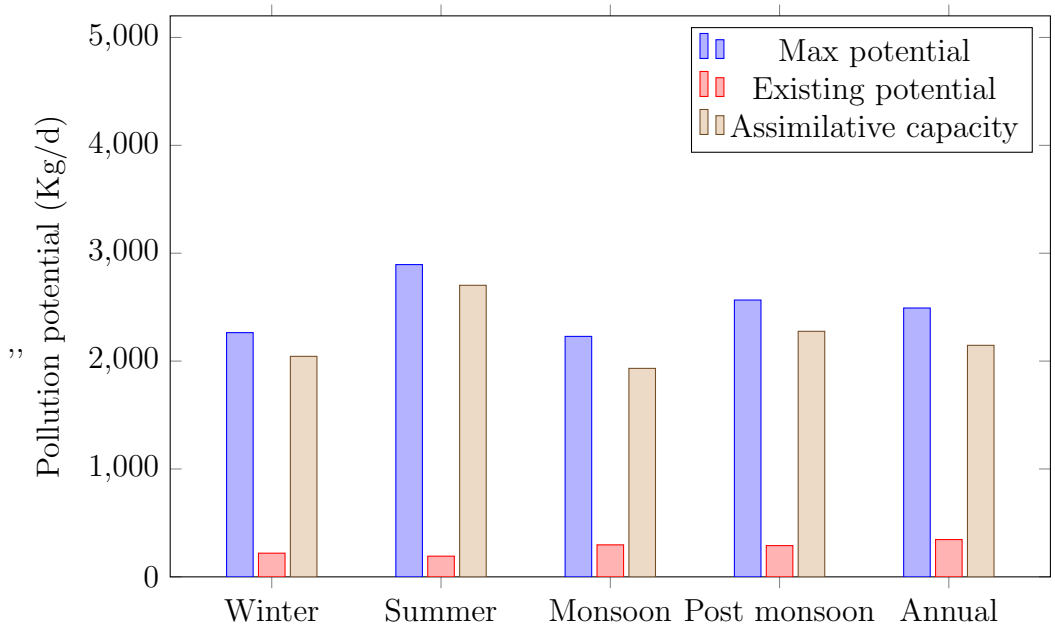
$$\text{Assimilative capacity} = \text{Max pollution potential - Existing pollution potential} \tag{5.2}$$

Seasonal variation of assimilative capacity shows that (fig 5.24)for NO_x monsoon has the lowest assimilative potential value of 1932.5 kg/d as it has significantly more existing pollution load of 296.7 kg/d as compared to other seasons but summer has the max assimilative potential value of 2702.6 kg/d as it has lowest existing pollution load of 191.8 kg/d from the industry to that of other seasons.

While for SO_2 it has been observed that monsoon has the lowest assimilative capacity value of 1964.7 kg/d though summer has the max available existing pollution potential value of 200 kg/d but surprisingly post monsoon has the max assimilative capacity of 3665 kg/d though winter has the lowest existing pollution potential of 124 kg/d.



(a) Assimilative capacity of SO_2



(b) Assimilative capacity of NO_x

Figure 5.24: Seasonal variation of assimilative capacity

Chapter 6

Findings and conclusion

6.1 Findings

6.1.1 Meteorological

- Max wind speed of 11.54 m/s was observed during summer for the entire year as shown in the fig 5.1b and the predominant wind direction is North east for the entire year as shown in the fig 5.1 and fig 5.2.
- In terms of seasonal average wind speed, summer has a max average value of 2.53 m/s and post monsoon has the lowest value of 1.56 m/s.
- Seasonal variation of temperature shows 5.4a monsoon is the hottest season with a seasonal average value of 28.53℃while average minimum temperature of 20.29℃prevails during winter.
- In terms of humidity max average humidity of 85.88% occurs during monsoon while minimum average humidity of 71.92% prevails during winter as shown in the fig 5.4b.
- Monthly and seasonal average variation of mixing height as shown in the fig 5.6 and fig 5.7 shows that evening hours have more mixing depths as compared to morning hours throughout the year.
- In terms of max average seasonal mixing height, a max value of of 654.2 meter during monsoon for morning hours and 825.8 meter during summer for evening hours respectively were observed.
- In terms of minimum seasonal average mixing height, a minimum value of 105.82 m for morning hours and 96.33 m for evening hours were observed during the winter.
- Seasonal variation of ventilation coefficient as shown in the fig 5.8, fig 5.9, fig 5.10 and fig 5.11 shows that for all the seasons evening hours have more ventilation coefficient value that to its morning hours.
- Average seasonal variation of ventilation coefficient for both morning and evening hours shows that, the area has low assimilative potential that indicates higher pollutant concentration.

- Daily variation of ventilation coefficient for both morning and evening hour shows the area has been within the low assimilative capacity zone for the entire year as max VC value of 5977 m^2/s is observed during 17th of March for evening hour that for morning hour the value reads 3899 m^2/s on 7th of April.

6.1.2 Dispersion modelling

- At existing pollution load for all the receptors (both uniform and discrete) the predicted GLS's are well below the NAAQS standard for both the pollutant and for all the seasons, indicating the area has a fair assimilative capacity.

- At existing pollution load monsoon is the most critical season as for NO_x max seasonal predicted GLC value of 10.7 $\mu g/m^3$ is observed and that for SO_2 second highest seasonal value of 6.05 $\mu g/m^3$ is observed during this season just behind the 6.3 $\mu g/m^3$ value of summer season.

- At existing pollution load among discrete receptors, railway station area has witnessed max pollutant concentration for all the seasons for both the pollutants as well and the same is applicable for max pollution load.

- Max assimilative capacity value of 2702.6 kg/d is observed during summer for NO_x and that for SO_2 a max value of 3665 kg/d is observed during post monsoon.

- At available assimilative pollution load monsoon season is experiencing the most critical periods as for both the pollutants it has the least pollutant potential value of 1932.5 kg/d for NO_x and 1964.7 kg/d for SO_2.

- At max pollutant potential at all the discrete receptors predicted GLC's are also well below the NAAQS standard.

6.2 Conclusion

Though monsoon season has the best average morning ventilation coefficient value and second best evening ventilation coefficient value just behind summer,also it is most humid as well as hottest season and second best average wind speed of 2.5 m/s failing by a mere gap of .03 m/s to summer - it is experiencing the most critical condition in terms of pollutant concentration for both existing pollution load as well as max assimilative pollution load. One of the main reason of this phenomena as monsoon has significantly high existing pollution load as compared to other seasons for both the pollutants and also the area has average ventilation coefficient ¡2000 m^2/s throughout the year indicating its low assimilative potential for all the seasons which can also be observed from the winter's assimilative potential value for both the pollutants which is very close to monsoon's value and also other two seasons are also not far ahead specially post monsoon's (2276 kg/d) NO_x assimilative value which is also pretty close to monsoon's (1932.5 kg/d) and similar thing is applicable to summer's (2335 kg/d) SO_2 assimilative value to monsoon's (1964.7 kg/d).

The above all scenario's clearly shows the low assimilative potential of the area though it has fair available assimilative potential at existing pollution load.

For all the seasons among discrete receptors max pollutant concentration has been observed near the railway station area, it is mainly due to as it is not only closer to the point sources also it is in the downwind direction. Because of this max predicted GLC's are observed in downwind direction (predominantly NE throughout the year) at a distance of 2.5 to 6.5 km, this is how wind direction has played a major role in the dispersion of pollutants over the study region.

At existing load and also at max pollution load among all the discrete receptors as well as over the entire study area max predicted GLC's of the pollutants are well below the NAAQS standard for both residential as well as non residential area.

So, to conclude we can say though the area has low assimilative potential, still it has a fair assimilative capacity and further air pollution load can be released on that region given that it will not violate the NAAQS standard.

6.3 Future scope of the study

- In this study only the emission from Digboi oil refinery is considered, hence emissions from other medium and small scale industries can also be considered.
- Vehicular traffic contributes a significant amount of pollution load as we have already seen from literature study, hence that can also be included for more accurate results.
- In Assam forest fire and domestic burning is a common phenomena and the Digboi area is also surrounded by village as well as forests hence inclusion of background source of pollution can also be done for better results.
- For the calculation of ventilation coefficient Holswat method is method is adopted the same can be done with the help of HYSPLIT model and a comparative analysis between these two results will give more accurate results.
- Here wind speed is taken at surface level average wind speed though it is actually the average wind speed at the level of mixing height which can be obtained from Sutton's power law.
- In this study only SO_2 and NO_x are considered and it can be further extended for particulate matter as it is a matter of grave concern for that area as well as CO as it is released not only from oil refinery but also the vehicular traffic and domestic burning releases a significant amount of CO into the atmosphere.

References

1. Afzali, A., Rashid, M., Afzali, M. & Younesi, V. Prediction of air pollutants concentrations from multiple sources using AERMOD coupled with WRF prognostic model. *Journal of Cleaner Production* **166,** 1216–1225 (2017).
2. Ashrafi, K., Shafipour, M. & Kamalan, H. Estimating temporal and seasonal variation of ventilation coefficients (2009).
3. Berman, S., Ku, J.-Y. & Rao, S. T. Spatial and temporal variation in the mixing depth over the northeastern United States during the summer of 1995. *Journal of Applied Meteorology* **38,** 1661–1673 (1999).
4. CPCB. National Ambient Air Quality standards. *Air quality standards.* https://https://cpcb.nic.in/uploads/National_Ambient_Air_Quality_Standards.pdf (2009).
5. Ghose, M. & Majee, S. Air pollution due to opencast coal mining and it's control in Indian context (2001).
6. Ghose, M. K. & Majee, S. Assessment of dust generation due to opencast coal mining–an Indian case study. *Environmental monitoring and assessment* **61,** 257–265 (2000).
7. Gibson, M. D., Kundu, S. & Satish, M. Dispersion model evaluation of PM2.5, NOx and SO2 from point and major line sources in Nova Scotia, Canada using AERMOD Gaussian plume air dispersion model. *Atmospheric Pollution Research* **4,** 157–167. https://www.sciencedirect.com/science/article/pii/S1309104215303858 (2013).
8. Goyal, P., Anand, S. & Gera, B. Assimilative capacity and pollutant dispersion studies for Gangtok city. *Atmospheric Environment* **40,** 1671–1682 (2006).
9. Goyal, S. & Rao, C. C. Air assimilative capacity-based environment friendly siting of new industries—A case study of Kochi region, India. *Journal of environmental management* **84,** 473–483 (2007).
10. Holzworth, G. C. Mixing depths, wind speeds and air pollution potential for selected locations in the United States1. *Journal of Applied Meteorology (1962-1982),* 1039–1044 (1967).
11. Huertas, J. I., Huertas, M. E. & Solıs, D. A. Characterization of airborne particles in an open pit mining region. *Science of the total Environment* **423,** 39–46 (2012).
12. Krishna, T. R., Reddy, M., Reddy, R. & Singh, R. Assimilative capacity and dispersion of pollutants due to industrial sources in Visakhapatnam bowl area. *Atmospheric Environment* **38,** 6775–6787 (2004).

13. Krishna, T. R., Reddy, M., Reddy, R. & Singh, R. Impact of an industrial complex on the ambient air quality: Case study using a dispersion model. *Atmospheric environment* **39,** 5395–5407 (2005).

14. Manju, N., Balakrishnan, R. & Mani, N. Assimilative capacity and pollutant dispersion studies for the industrial zone of Manali. *Atmospheric Environment* **36,** 3461–3471 (2002).

15. NEERI. The National Environment Engineering Research Institute. *Zones of india based on ventilation coefficient.* https://www.neeri.res.in/#googtrans(en%7Cen) (1990).

16. Panda, S. & Nagendra, S. S. Assimilative capacity–based emission load management in a critically polluted industrial cluster. *Journal of the Air & Waste Management Association* **67,** 1353–1363 (2017).

17. Prakash, B., Majumder, S., Swamy, M. & Mahesh, S. Assimilative capacity and air quality index studies of the atmosphere in Hebbal industrial area, Mysuru. *International Journal of Innovative Research in Science, Engineering and Technology* **6,** 20651–20662 (2017).

18. Shaikh, M. SPM assimilative capacity assessment of MundraTaluka. *Indian Journal of Air Pollution Control* **10,** 68–77 (2010).

19. Singh, G. & Perwez, A. Estimation of assimilative capacity of the airshed in iron ore mining region of Goa. *Indian Journal of Science and Technology* **8,** 1 (2015).

20. Sturm, P.-J., Hausberger, S., Keller, M. & Haan, P. d. Estimating real world emissions from passenger cars-use and limitations of instantaneous emission data. *International Journal of Vehicle Design* **24,** 19–33 (2000).

21. Thawonkaew, A., Thepanondh, S., Sirithian, D. & Jinawa, L. Assimilative capacity of air pollutants in an area of the largest petrochemical complex in Thailand. *GEOMATE Journal* **11,** 2162–2169 (2016).